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DATA-PARALLEL VECTOR TYPES & OPERATIONS

ABSTRACT

This paper describes class templates for portable data-parallel (e.g. SIMD) programming via vector types.

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0

REMARKS

- This documents talks about “vector” types/objects. In general this will not refer to the `std::vector` class template. References to the container type will explicitly call out the `std` prefix to avoid confusion.
- In the following, $\mathcal{W}_T$ denotes the number of scalar values (width) in a vector of type T (sometimes also called the number of SIMD lanes)
- [N4184], [N4185], and [N4395] provide more information on the rationale and design decisions. [N4454] discusses a matrix multiplication example. My PhD thesis [1] contains a very thorough discussion of the topic.
- This paper is not supposed to specify a complete API for data-parallel types and operations. It is meant as a useful starting point. Once the foundation is settled on, higher level APIs will be proposed.

1

CHANGELOG

1.1

CHANGES FROM R8

Previous revision: [P0214R8].

- Use the word-of-power “width” in the specification of `size()`.
- Aligned wording in `copy_from` and `copy_to` restricting pointer alignment with the wording in `[ptr.align]`. Also added wording for the meaning of `element_aligned_tag`.
- Fixed missing `SFINAE` constraints on `Flags` in masked load. Streamlined the wording to use itemization for listing multiple constraints.
- Removed `const` from the second template argument to `const_where_expression`. Consequently, removed `remove_const_t` from the definitions and wording in `const_where_expression`. (I verified the change in my implementation.)
- Use “extended ABI tag” instead of “implementation-defined ABI tag”. The corresponding word of power is defined in `[simd.abi]`.
- Renamed `abi_for_size<T, N>` to `simd_abi::deduce<T, N>`.
- Improved wording in case multiple ABI tags match the constraints passed to `simd_abi::deduce<T, N>`.

- Clarify valid values for `N` for `is_simd_flag_type<overaligned_tag<N>>` ([simd.traits]).
- Call out that user specialization of traits is UB.
- Simplified `memory_alignment` SFINAE logic.
- Improved wording for selected elements/indices.
- Rewrite the logic for what template arguments `simd` and `simd_mask` are supported.
- Add an example to [simd.overview].
- Handle cv-qualification in broadcast ctor ([simd.ctor]).
- Call out constraints on the generator callable in the generator ctor ([simd.ctor]).
- Add Throws specification to `reduce` overloads that a user-defined `binary_op`.
- Reword `simd::reference` using an exposition-only class definition ([simd.reference]).

1.2

CHANGES FROM R7

Previous revision: [P0214R7].

- Removed the explicit signatures of the special math functions in favor of a rule to translate the `<cmath>` signatures to `simd` types. This resolves an over-specification, which disallowed certain valid implementation strategies. (see [simd.math])
- Added margin-note questioning non-normative note about special math function preconditions. (see [simd.math])
- Fixed and finalized note on the intent of the data-parallel library. (see [simd.general])
- Renamed `neutral_element` to `identity_element` and specified the requirement on its value in relation to `binary_op`. Removed the note explaining an implementation detail. (see [simd.reductions])

1.3

CHANGES FROM R6

Previous revision: [P0214R6].

- Fixed return type of `isinf`. ([simd.math])
- ✓ Changed namespace to `std::experimental::parallelism_v2`.
- ✓ Shortened intro sentence in [simd.syn].
- ✓ Added missing `const_where_expression` overload for `where(bool, const T&)`. ([simd.syn], [simd.mask.where]) (The overload is present in my reference implementation.)
- ✓ Removed `const` from `const_where_expression` return type. ([simd.syn])
 - Fixed incorrect `const where_expression` return types in [simd.mask.where].
- ✓ Removed `flags` namespace [simd.syn].
- ✓ all of your global `constexpr` variables should be `inline`
- ✓ Rename template parameters `A` to `Abis` and `As...` to `Abis....`
- ✓ Stable names follow a hierarchy; use [simd.class] instead of [simd]; rename [simd_mask] to [simd.mask]
- ✓ `reduce`: Moved defaulted `BinaryOperation` argument to the end.
- ✓ `reduce`: Replaced `default_natural_element` with *implementation-defined*
- ✓ `reduce`: Explained what the default `natural_element` is supposed to do. ([simd.reductions])
- ✓ Consistently initialize tags.
- ✓ Put the “see belows” in italics instead of comments.
- ✓ Consistently pass by const-ref. (LWG: Why are the arguments by value? Be consistent. Resolved to consistently pass by const-ref. Rationale: if an implementation wants a non-inline, pass-by-value function it can inline the public const-ref function and call an internal function. If the public function uses by value parameters it may happen that the implied copy is not optimized away.)
- ✓ Simplify casts using aliases or default arguments.

- ✓ Split second `reduce` overload in six functions, one without default `neutral_element` and five explicitly overloaded for `plus<>`, `multiplies<>`, `bit_and<>`, `bit_or`, and `bit_xor`.
- ✓ Improved `requires` clause on `reduce` to mention return type and generalize ABI tag type requirement.
- ✓ Require integral `value_type` for bitwise reductions.
- ✓ Removed impossible condition from `simd_cast` remark.
- ✓ Fixed masked `hmin` and `hmax` specification to use “forallmasked”.
- ✓ Forward `min`, `max`, `minmax`, and `clamp` to `std::min/max/minmax/clamp`.
- ✓ `scalar` is not an alias for `fixed_size<1>`.
- ✓ “an implementation *shall* support at least *all* ...” ([simd.abi])
- ✓ Replaced “exact-bool” arguments from *implementation-defined* to *see below*.
- ✓ Reversed “if” to “unless” logic.
- ✓ Fixed `is_mask` to `is_simd_mask`.
- ✓ Fixed incorrect `simd_mask` exposition-only member name to `mask` ([simd.where-expr]).
- ✓ Fixed constraints on generator ctor to require generator be callable with all element indices.
- ✓ Fixed wording to allow vectorized execution of the generator.
- ✓ Moved all wording about “target” or “architecture” into non-normative notes.
- ✓ Add `is_simd_flag_type` trait and use it for all loads and stores.
- ✓ Define and use the term *vectorizable type* to simplify the wording.
- ✓ Define “selected elements” in `where_expression` to use it instead of `data[i]` where `mask[i]` is `true`.
- ✓ [simd.whereexpr] reword what the members of `where_expression` mean and where they come from
- ✓ Replaced “floating-point and integral” with “arithmetic”.

- ✓ Consistently use “element” instead of “component”.
- ✓ Consistently place “and”/“or” and the end of bullet points instead of the front.
 - Allow shifts on signed integers as was requested by LEWG.
- ✓ Added introduction in [simd.general].
- ✓ Added wording in [simd.general] to clarify that the application of operations/-operators on elements in `simd` and `simd_mask` are unsequenced with respect to each other.
- ✓ Fix Order: requires, effects, sync, post-cond, returns, throws, complexity, remarks, error-cond
- ✓ Removed repetitions of clause names.
- ✓ “Let X be foo” doesn’t need a “Remarks” clause.
- ✓ Defined “vectorizable types” as a term for “arithmetic types other than bool”.
 - Removed immutable masked load. Requested in LWG review session because it’s too clever and may block/hinder acceptance. LEWG queried/informed via reflector. Quick review of the issue at the start of Jacksonville meeting requested.
 - Introduced exposition-only `nodeduce_t` to replace `remove_const_t` ([simd.mask.where]).
 - Removed `noexcept` from `hmin` and `hmax` in the synopsis, to match the declarations below.

1.4

CHANGES FROM R5

Previous revision: [P0214R5].

- Renamed `memload` to `copy_from` and `memstore` to `copy_to`. ([simd.copy], [simd.mask.copy])
- Fixed `split` to never convert the `value_type`. ([simd.casts])
- Added missing `long double` overload of `ceil`. ([simd.math])

1.5

CHANGES FROM R4

Previous revision: [P0214R4].

- Changed `align_val_t` argument of `overaligned<N>` and `overaligned_tag<N>` to `size_t`. (Usage is otherwise too cumbersome.)

CHANGES AFTER LEWG REVIEW

- Remove section on naming (after the topic was discussed and decided in LEWG).
- Rename `datapar` to `simd` and `mask` to `simd_mask`.
- Remove incorrect template parameters to scalar boolean reductions.
- Merge proposed `simd_cast` into the wording (using Option 3) and extend `static_simd_cast` accordingly.
- Merge proposed `split` and `concat` functions into the wording.
- Since the target is a TS, place headers into `experimental/` directory.

1.6

CHANGES FROM R3

Previous revision: [P0214R3].

CHANGES BEFORE KONA

- Add special math overloads for signed char and short. They are important to avoid widening to multiple SIMD registers and since no integer promotion is applied for `simd` types.
- Editorial: Prefer `using` over `typedef`.
- Overload shift operators with `int` argument for the right hand side. This enables more efficient implementations. This signature is present in the Vc library, and was forgotten in the wording.
- Remove empty section about the omission of logical operators.
- Modify `simd_mask` compares to return a `simd_mask` instead of `bool` ([`simd.mask.comparison`]). This resolves an inconsistency with all the other binary operators.
- Editorial: Improve `reference` member specification ([`simd.overview`]).
- Require `swap(v[0], v[1])` to be valid ([`simd.overview`]).

- Fixed inconsistency of masked load constructor after move of `memload` to `where_expression` ([`simd.whereexpr`]).
- Editorial: Use `Requires` clause instead of `Remarks` to require the memory argument to loads and stores to be large enough ([`simd.whereexpr`], [`simd.copy`], [`simd.mask.copy`]).
- Add a note to special math functions that precondition violation is UB ([`simd.math`]).
- Bugfix: Binary operators involving two `simd::reference` objects must work ([`simd.overview`]).
- Editorial: Replace `Note` clauses in favor of [*Note: — end note*].
- Editorial: Replace UB `Remarks` on load/store alignment requirements with `Requires` clauses.
- Add an example section (4).

CHANGES AFTER KONA

— design related:

- Readd `bool` overloads of mask reductions and ensure that implicit conversions to `bool` are ill-formed.
- Clarify effects of using an ABI parameter that is not available on the target ([`simd.overview`], [`simd.mask.overview`], [`simd.traits`]).
- Split `where_expression` into `const` and `non-const` class templates.
- Add section on naming.
- Discuss the questions/issues raised on `max_fixed_size` in Kona (Section 7.11).
- Make `max_fixed_size` dependent on `T`.
- Clarify that converting loads and stores only work with arrays of non-bool arithmetic type ([`simd.copy`]).
- Discuss `simd_mask` and `bitset` reduction interface differences.
- Relax requirements on return type of generator function for the generator constructor ([`simd.ctor`]).

- Remove overly generic `simd_cast` function.
- Add proposal for a widening cast function.
- Add proposal for `split` and `concat` cast functions.
- Add `noexcept` or “Throws: Nothing.” to most functions.

— wording fixes & improvements:

- Remove non-normative noise about ABI tag types ([`simd.abi`]).
- Remove most of the text about vendor-extensions for ABI tag types, since it’s QoI ([`simd.abi`]).
- Clarify the differences and intent of `compatible<T>` vs. `native<T>` ([`simd.abi`]).
- Move definition of `where_expression` out of the synopsis ([`simd.whereexpr`]).
- Editorial: Improve `is_simd` and `is_mask` wording ([`simd.traits`]).
- Make *ABI tag* a consistent term and add `is_abi_tag` trait ([`simd.traits`], [`simd.abi`]).
- Clarify that `simd_abi::fixed_size<N>` must support all `N` matching all possible extended ABI tags ([`simd.abi`]).
- Clarify `abi_for_size` wording ([`simd.traits`]).
- Turn `memory_alignment` into a trait with a corresponding `memory_alignment_v` variable template.
- Clarify `memory_alignment` wording; when it has no `value` member; and imply its value through a reference to the load and store functions ([`simd.traits`]).
- Remove exposition-only `where_expression` constructor and make exposition-only data members private ([`simd.whereexpr`]).
- Editorial: use “shall not participate in overload resolution unless” consistently.
- Add a note about variability of `max_fixed_size` ([`simd.abi`]).
- Editorial: use “target architecture” and “currently targeted system” consistently.
- Add margin notes presenting a wording alternative that avoids “target system” and “target architecture” in normative text.

- Specify result of masked reduce with empty mask ([simd.reductions]).
- Editorial: clean up the use of “supported” and resolve contradictions resulting from incorrect use of conventions in the rest of the standard text ([simd.overview], [simd.mask.overview], [simd.traits]).
- Add Section 6 Feature Detection Macros.

1.7

CHANGES FROM R2

Previous revision: [P0214R2].

- Fixed return type of masked `reduce` ([simd.reductions]).
- Added binary `min`, `max`, `minmax`, and `clamp` ([simd.alg]).
- Moved member `min` and `max` to non-member `hmin` and `hmax`, which cannot easily be optimized from `reduce`, since no function object such as `std::plus` exists ([simd.reductions]).
- Fixed neutral element of masked `hmin/hmax` and drop UB ([simd.reductions]).
- Removed remaining reduction member functions in favor of non-member `reduce` (as requested by LEWG).
- Replaced `init` parameter of masked `reduce` with `neutral_element` ([simd.reductions]).
- Extend `where_expression` to support `const simd` objects ([simd.mask.where]).
- Fixed missing `explicit` keyword on `simd_mask(bool)` constructor ([simd.mask.ctor]).
- Made binary operators for `simd` and `simd_mask` friend functions of `simd` and `simd_mask`, simplifying the SFINAE requirements considerably ([simd.binary], [simd.mask.binary]).
- Restricted broadcasts to only allow non-narrowing conversions ([simd.ctor]).
- Restricted `simd` to `simd` conversions to only allow non-narrowing conversions with `fixed_size` ABI ([simd.ctor]).
- Added generator constructor (as discussed in LEWG in Issaquah) ([simd.ctor]).
- Renamed `copy_from` to `memload` and `copy_to` to `memstore`. ([simd.copy], [simd.mask.copy])

- Documented effect of `overaligned_tag<N>` as `Flags` parameter to `load/store`. ([simd.copy], [simd.mask.copy])
- Clarified cv requirements on `T` parameter of `simd` and `simd_mask`.
- Allowed all implicit `simd_mask` conversions with `fixed_size` ABI and equal size ([simd.mask.ctor]).
- Made increment and decrement of `where_expression` return `void`.
- Added `static_simd_cast` for simple casts ([simd.casts]).
- Clarified default constructor ([simd.overview], [simd.overview]).
- Clarified `simd` and `simd_mask` with invalid template parameters to be complete types with deleted constructors, destructor, and assignment ([simd.overview], [simd.overview]).
- Wrote a new subsection for a detailed description of `where_expression` ([simd.where-expr]).
- Moved masked loads and stores from `simd` and `simd_mask` to `where_expression` ([simd.whereexpr]). This required two more overloads of `where` to support value objects of type `simd_mask` ([simd.mask.where]).
- Removed `where_expression::operator!` ([simd.whereexpr]).
- Added aliases `native_simd`, `native_mask`, `fixed_size_simd`, `fixed_size_mask` ([simd.syn]).
- Removed `bool` overloads of mask reductions awaiting a better solution ([simd.mask.reductions]).
- Removed special math functions with `f` and `l` suffix and `l` and `ll` prefix ([simd.math]).
- Modified special math functions with mixed types to use `fixed_size` instead of `abi_for_size` ([simd.math]).
- Added simple ABI cast functions `to_fixed_size`, `to_native`, and `to_compatible` ([simd.casts]).

1.8

CHANGES FROM R1

Previous revision: [P0214R1].

- Fixed converting constructor synopsis of `simd` and `simd_mask` to also allow varying Abi types.
- Modified the wording of `simd_mask::native_handle()` to make the existence of the functions implementation-defined.
- Updated the discussion of member types to reflect the changes in R1.
- Added all previous SG1 straw poll results.
- Fixed *commonabi* to not invent native Abi that makes the operator ill-formed.
- Dropped table of math functions.
- Be more explicit about the implementation-defined Abi types.
- Discussed resolution of the `simd_abi::fixed_size<N>` design (7.7.4).
- Made the `compatible` and `native` ABI aliases depend on `T` ([`simd.abi`]).
- Added `max_fixed_size` constant ([`simd.abi`]).
- Added masked loads.
- Added rationale for return type of `simd::operator-()` (7.10).

— SG1 guidance:

- Dropped the default load / store flags.
- Renamed the (un)aligned flags to `element_aligned` and `vector_aligned`.
- Added an `overaligned<N>` load / store flag.
- Dropped the ampersand on `native_handle` (no strong preference).
- Completed the set of math functions (i.e. add trig, log, and exp).

— LEWG (small group) guidance:

- Dropped `native_handle` and add non-normative wording for supporting `static_cast` to implementation-defined SIMD extensions.

- Dropped non-member load and store functions. Instead have `copy_from` and `copy_to` member functions for loads and stores. ([`simd.copy`], [`simd.mask.copy`]) (Did not use the `load` and `store` names because of the unfortunate inconsistency with `std::atomic`.)
- Added algorithm overloads for `simd` reductions. Integrate with `where` to enable masked reductions. ([`simd.reductions`]) This made it necessary to spell out the class `where_expression`.

1.9

CHANGES FROM R0

Previous revision: [P0214R0].

- Extended the `simd_abi` tag types with a `fixed_size<N>` tag to handle arbitrarily sized vectors ([`simd.abi`]).
- Converted `memory_alignment` into a non-member trait ([`simd.traits`]).
- Extended implicit conversions to handle `simd_abi::fixed_size<N>` ([`simd.ctor`]).
- Extended binary operators to convert correctly with `simd_abi::fixed_size<N>` ([`simd.binary`]).
- Dropped the section on “`simd` logical operators”. Added a note that the omission is deliberate.
- Added logical and bitwise operators to `simd_mask` ([`simd.mask.binary`]).
- Modified `simd_mask` compares to work better with implicit conversions ([`simd.mask.comparison`]).
- Modified `where` to support different Abi tags on the `simd_mask` and `simd` arguments ([`simd.mask.where`]).
- Converted the load functions to non-member functions. SG1 asked for guidance from LEWG whether a load-expression or a template parameter to load is more appropriate.
- Converted the store functions to non-member functions to be consistent with the load functions.
- Added a note about masked stores not invoking out-of-bounds accesses for masked-off elements of the vector.

- Converted the return type of `simd::operator[]` to return a smart reference instead of an lvalue reference.
- Modified the wording of `simd_mask::operator[]` to match the reference type returned from `simd::operator[]`.
- Added non-trig/pow/exp/log math functions on `simd`.
- Added discussion on defaulting load/store flags.
- Added sum, product, min, and max reductions for `simd`.
- Added load constructor.
- Modified the wording of `native_handle()` to make the existence of the functions implementation-defined, instead of only the return type. Added a section in the discussion (cf. Section 7.8).
- Fixed missing flag objects.

2

STRAW POLLS

2.1

SG1 AT CHICAGO 2013

Poll: Pursue SIMD/data parallel programming via types?

SF	F	N	A	SA
1	8	5	0	0

2.2

SG1 AT URBANA 2014

- Poll: SF = ABI via namespace, SA = ABI as template parameter

SF	F	N	A	SA
0	0	6	11	2

- Poll: Apply size promotion to vector operations? SF = `shortv + shortv = intv`

SF	F	N	A	SA
1	2	0	6	11

- Poll: Apply “sign promotion” to vector operations? SF = `ushortv + shortv = ushortv`; SA = no mixed signed/unsigned arithmetic

SF	F	N	A	SA
1	5	5	7	2

2.3

SG1 AT LENEXA 2015

Poll: Make vector types ready for LEWG with arithmetic, compares, write-masking, and math?

SF	F	N	A	SA
10	6	2	0	0

2.4

SG1 AT JAX 2016

- Poll: Should subscript operator return an lvalue reference?

SF	F	N	A	SA
0	6	10	2	1

- Poll: Should subscript operator return a “smart reference”?

SF	F	N	A	SA
1	7	10	0	0

- Poll: Specify simd width using ABI tag, with a special template tag for fixed size.

SF	F	N	A	SA
3	7	0	0	1

- Poll: Specify simd width using <T, N, abi>, where abi is not specified by the user.

SF	F	N	A	SA
1	2	5	2	1

2.5

SG1 AT OULU 2016

- Poll: Keep `native_handle` in the wording (dropping the ampersand in the return type)?

SF	F	N	A	SA
0	6	3	3	0

- Poll: Should the interface provide a way to specify a number for over-alignment?

SF	F	N	A	SA
2	6	5	0	0

- Poll: Should loads and stores have a default load/store flag?

SF	F	N	A	SA
0	0	7	4	1

2.6

LEWG AT ISSAQUAH 2016

- Poll: Unary minus on unsigned simd should be ill formed

SF	F	N	A	SA
0	5	6	0	3

- Poll: Reductions only as free functions
→ unanimous consent
- Poll: Jens should work with the author and return with an updated paper
→ unanimous consent

2.7

LEWG AT KONA 2017

- Poll: Want `operator<<(signed)` to work except where it's undefined for the underlying integer?
→ unanimous consent
- Poll: Should there be overloads for `where` and `(simd_mask and simd)` reductions with `bool/builtin` types in place of `simd_mask/simd`?

SF	F	N	A	SA
1	5	6	1	0

- Poll: Should there be a named “widen” function that widens the element type `T` from (e.g.) `int` to `long`, but rejects `int` to `short`? The number of elements is not changed.
→ unanimous consent
- Poll: Should there be a named “concat” functions that concatenates several `simds` with the same element type, but potentially different length?
The latter two operations are currently lumped together in `simd_cast`.
The widen / concat questions, I think, comes down to whether we want the current `simd_cast` or not.
→ unanimous consent
- Poll: We still have a few open ends on implicit conversions (<https://github.com/mattkretz/wg21-papers/issues/26>, <https://github.com/mattkretz/wg21-papers/issues/3>). The current paper is rather strict, there may be room for more implicit conversions without introducing safety problems. Should we postpone any changes in this area to after TS feedback?
→ unanimous consent

- Poll: Keep or drop `simd_abi::max_fixed_size`? (cf. <https://github.com/mattkretz/wg21-papers/issues/38>)

Authors will come back with a proposal

cf. Section 7.11

- Poll: Add a list of possible names for “simd” and “where” to the paper.
→ unanimous consent

2.8

LEWG AT TORONTO 2017

- Poll: Use Option 3 (support both element type and `simd` type as cast template argument, instead of just either one of them) for `static_simd_cast` and `simd_cast`.

→ unanimous consent

- Poll: Use `abi_for_size_t` instead of unconditionally using `fixed_size` for `split` and `concat`. Revisit the question after TS feedback.

→ unanimous consent

- Poll: Name it `datapar` (in favor) or `simd` (against)?

SF	F	N	A	SA
1	4	1	5	4

We want TS feedback on the name, i.e. whether anyone is confused about the applicability of the feature because of the name.

- Poll: Name `mask` (in favor) or `simdmask` (against)?

SF	F	N	A	SA
0	1	2	5	7

- Poll: Name `simd_mask` (in favor) or `simdmask` (against)?

SF	F	N	A	SA
4	6	3	1	0

- Poll: `where` (in favor) vs. `masked` (against)?

SF	F	N	A	SA
5	1	3	2	3

No consensus for change, revisit after TS feedback.

- Vote for your preference for a name for the load/store functions

load	memload	load_from	copy_from
4	2	6	7

- Poll: `load_from` (in favor) vs. `copy_from` (against)?

SF	F	N	A	SA
2	3	3	0	3

No consensus to override the author's preference (`copy_from`).

- Poll: Keep `simd_mask` reductions as free functions with the current names. Consider follow-up papers if any names need to be changed (e.g. for consistency).
→ unanimous consent

- Poll: Forward to LWG for TS

SF	F	N	A	SA
11	2	0	0	0

2.9

LEWG AT JACKSONVILLE 2018

- Poll: Change the parameter order for `simd reduce` to place the identity value after the binary op?

SF	F	N	A	SA
0	0	2	10	3

- Poll: Change `std::abi_for_size_t` to `simd_abi::deduce_t`?

SF	F	N	A	SA
2	8	1	0	0

3

INTRODUCTION

3.1

SIMD REGISTERS AND OPERATIONS

Since many years the number of SIMD instructions and the size of SIMD registers have been growing. Newer microarchitectures introduce new operations for optimizing certain (common or specialized) operations. Additionally, the size of SIMD registers has increased and may increase further in the future.

The typical minimal set of SIMD instructions for a given scalar data type comes down to the following:

- Load instructions: load $\mathcal{W}_T$ successive scalar values starting from a given address into a SIMD register.
- Store instructions: store from a SIMD register to $\mathcal{W}_T$ successive scalar values at a given address.
- Arithmetic instructions: apply the arithmetic operation to each pair of scalar values in the two SIMD registers and store the results back to a SIMD register.
- Compare instructions: apply the compare operation to each pair of scalar values in the two SIMD registers and store the results back to a SIMD mask register.
- Bitwise instructions: bitwise operations on SIMD registers.
- Shuffle instructions: permutation and/or blending of scalars in (a) SIMD register(s).

The set of available instructions may differ considerably between different microarchitectures of the same CPU family. Furthermore there are different SIMD register sizes. Future extensions will certainly add more instructions and larger SIMD registers.

3.2

MOTIVATION FOR DATA-PARALLEL TYPES

SIMD registers and operations are the low-level ingredients to efficient programming for SIMD CPUs. At a more abstract level this is not only about SIMD CPUs, but efficient data-parallel execution (CPUs, GPUs, possibly FPGAs and classical vector supercomputers). Operations on fundamental types in C++ form the abstraction for CPU registers and instructions. Thus, a data-parallel type (SIMD type) can provide the necessary interface for writing software that can utilize data-parallel hardware efficiently. Higher-level abstractions can be built on top of these types. Note that if a low-level access to SIMD is not provided, users of C++ are either constrained to work within the limits of the provided abstraction or resort to non-portable extensions, such as SIMD intrinsics.

In some cases the compiler might generate better code if only the intent is stated instead of an exact sequence of operations. Therefore, higher-level abstractions might seem preferable to low-level SIMD types. In my experience this is a non-issue because programming with SIMD types makes intent very clear and compilers can optimize sequences of SIMD operations just like they can for scalar operations. SIMD types do not lead to an easy and obvious answer for efficient and easily usable data structures, though. But, in contrast to vector loops, SIMD types make unsuitable data

structures glaringly obvious and can significantly support the developer in creating more suitable data layouts.

One major benefit from SIMD types is that the programmer can gain an intuition for SIMD. This subsequently influences further design of data structures and algorithms to better suit SIMD architectures.

There are already many users of SIMD intrinsics (and thus a primitive form of SIMD types). Providing a cleaner and portable SIMD API would provide many of them with a better alternative. Thus, SIMD types in C++ would capture and improve on widespread existing practice.

The challenge remains in providing *portable* SIMD types and operations.

3.3

PROBLEM

C++ has no means to use SIMD operations directly. There are indirect uses through automatic loop vectorization or optimized algorithms (that use extensions to C/C++ or assembly for their implementation).

All compiler vendors (that I worked with) add intrinsics support to their compiler products to make SIMD operations accessible from C. These intrinsics are inherently not portable and most of the time very directly bound to a specific instruction. (Compilers are able to statically evaluate and optimize SIMD code written via intrinsics, though.)

4

EXAMPLES

4.1

LOOP VECTORIZATION

This shows a low-level approach of manual loop chunking + epilogue for vectorization (“Leave no room for a lower-level language below C++ (except assembler).” [2]). It also shows SIMD loads, operations, write-masking (blending), and stores.

```
using floatv = native_simd<float>;
void f() {
    alignas(memory_alignment_v<floatv>) float data[N];
    fill_data(data);
    size_t i = 0;
    for (; i + floatv::size() <= N; i += floatv::size()) {
        floatv v(&data[i], vector_aligned);
        where(v > 100.f, v) = 100.f + (v - 100.f) * 0.1f;
        v.copy_to(&data[i], vector_aligned);
    }
    for (; i < N; ++i) {
        float x = data[i];
```

```

    if (x > 100.f) {
        x = 100.f + (x - 100.f) * 0.1f;
    }
    data[i] = x;
}
}

```

5

WORDING

The following is a draft targeting inclusion into the Parallelism TS 2. It defines a basic set of data-parallel types and operations.

(5.1) 8 Data-Parallel Types [simd.types]

(5.1.1) 8.1 General [simd.general]

- 1 The data-parallel library consists of data-parallel types and operations on these types. A data-parallel type consists of elements of an underlying arithmetic type, called the *element type*. The number of elements is a constant for each data-parallel type and called the *width* of that type.
- 2 Throughout this Clause, the term *data-parallel type* refers to all *supported* [simd.overview] specializations of the `simd` and `simd_mask` class templates. A *data-parallel object* is an object of *data-parallel type*.
- 3 An *element-wise operation* applies a specified operation to the elements of one or more data-parallel objects. Each such application is unsequenced with respect to the others. A *unary element-wise operation* is an element-wise operation that applies a unary operation to each element of a data-parallel object. A *binary element-wise operation* is an element-wise operation that applies a binary operation to corresponding elements of two data-parallel objects.
- 4 Throughout this Clause, the set of *vectorizable types* for a data-parallel type comprises all cv-unqualified arithmetic types other than `bool`.
- 5 [*Note:* The intent is to support acceleration through data-parallel execution resources, such as SIMD registers and instructions or execution units driven by a common instruction decoder. If such execution resources are unavailable, the interfaces support a transparent fallback to sequential execution. — *end note*]

(5.1.2) 8.2 Header `<experimental/simd>` synopsis [simd.syn]

```

namespace std::experimental {
    inline namespace parallelism_v2 {
        namespace simd_abi {
            struct scalar {};
            template <int N> struct fixed_size {};
            template <typename T> inline constexpr int max_fixed_size = implementation-defined;
            template <typename T> using compatible = implementation-defined;
            template <typename T> using native = implementation-defined;

            template <class T, size_t N> struct deduce { using type = see below; };
        }
    }
}

```

```

    template <class T, size_t N> using deduce_t = typename deduce<T, N>::type;
}

struct element_aligned_tag {};
struct vector_aligned_tag {};
template <size_t> struct overaligned_tag {};
inline constexpr element_aligned_tag element_aligned{};
inline constexpr vector_aligned_tag vector_aligned{};
template <size_t N> inline constexpr overaligned_tag<N> overaligned{};

// traits [simd.traits]
template <class T> struct is_abi_tag;
template <class T> inline constexpr bool is_abi_tag_v = is_abi_tag<T>::value;

template <class T> struct is_simd;
template <class T> inline constexpr bool is_simd_v = is_simd<T>::value;

template <class T> struct is_simd_mask;
template <class T> inline constexpr bool is_simd_mask_v = is_simd_mask<T>::value;

template <class T> struct is_simd_flag_type;
template <class T> inline constexpr bool is_simd_flag_type_v = is_simd_flag_type<T>::value;

template <class T, class Abi = simd_abi::compatible<T>> struct simd_size;
template <class T, class Abi = simd_abi::compatible<T>>
inline constexpr size_t simd_size_v = simd_size<T, Abi>::value;

template <class T, class U = typename T::value_type> struct memory_alignment;
template <class T, class U = typename T::value_type>
inline constexpr size_t memory_alignment_v = memory_alignment<T, U>::value;

// class template simd [simd.class]
template <class T, class Abi = simd_abi::compatible<T>> class simd;
template <class T> using native_simd = simd<T, simd_abi::native<T>>;
template <class T, int N> using fixed_size_simd = simd<T, simd_abi::fixed_size<N>>;

// class template simd_mask [simd.mask.class]
template <class T, class Abi = simd_abi::compatible<T>> class simd_mask;
template <class T> using native_simd_mask = simd_mask<T, simd_abi::native<T>>;
template <class T, int N> using fixed_size_simd_mask = simd_mask<T, simd_abi::fixed_size<N>>;

// casts [simd.casts]
template <class T, class U, class Abi> see below simd_cast(const simd<U, Abi>&);
template <class T, class U, class Abi> see below static_simd_cast(const simd<U, Abi>&);

template <class T, class Abi>
fixed_size_simd<T, simd_size_v<T, Abi>> to_fixed_size(const simd<T, Abi>&) noexcept;
template <class T, class Abi>
fixed_size_simd_mask<T, simd_size_v<T, Abi>> to_fixed_size(const simd_mask<T, Abi>&) noexcept;
template <class T, int N> native_simd<T> to_native(const fixed_size_simd<T, N>&) noexcept;
template <class T, int N> native_simd_mask<T> to_native(const fixed_size_simd_mask<T, N>&) noexcept;
template <class T, int N> simd<T> to_compatible(const fixed_size_simd<T, N>&) noexcept;
template <class T, int N> simd_mask<T> to_compatible(const fixed_size_simd_mask<T, N>&) noexcept;

template <size_t... Sizes, class T, class Abi>
tuple<simd<T, simd_abi::deduce_t<T, Sizes>>...> split(const simd<T, Abi>&);
template <size_t... Sizes, class T, class Abi>
tuple<simd_mask<T, simd_abi::deduce_t<T, Sizes>>...> split(const simd_mask<T, Abi>&);

```

```

template <class V, class Abi>
array<V, simd_size_v<typename V::value_type, Abi> / V::size()> split(
    const simd<typename V::value_type, Abi>&);
template <class V, class Abi>
array<V, simd_size_v<typename V::value_type, Abi> / V::size()> split(
    const simd_mask<typename V::value_type, Abi>&);

template <class T, class... Abis>
simd<T, simd_abi::deduce_t<T, (simd_size_v<T, Abis> + ...) >> concat(const simd<T, Abis>&...);
template <class T, class... Abis>
simd_mask<T, simd_abi::deduce_t<T, (simd_size_v<T, Abis> + ...) >> concat(const simd_mask<T, Abis>&...);

// reductions [simd.mask.reductions]
template <class T, class Abi> bool all_of(const simd_mask<T, Abi>&) noexcept;
template <class T, class Abi> bool any_of(const simd_mask<T, Abi>&) noexcept;
template <class T, class Abi> bool none_of(const simd_mask<T, Abi>&) noexcept;
template <class T, class Abi> bool some_of(const simd_mask<T, Abi>&) noexcept;
template <class T, class Abi> int popcount(const simd_mask<T, Abi>&) noexcept;
template <class T, class Abi> int find_first_set(const simd_mask<T, Abi>&);
template <class T, class Abi> int find_last_set(const simd_mask<T, Abi>&);

bool all_of(see below) noexcept;
bool any_of(see below) noexcept;
bool none_of(see below) noexcept;
bool some_of(see below) noexcept;
int popcount(see below) noexcept;
int find_first_set(see below) noexcept;
int find_last_set(see below) noexcept;

// masked assignment [simd.whereexpr]
template <class M, class T> class const_where_expression;
template <class M, class T> class where_expression;

// masked assignment [simd.mask.where]
template <class T> struct nodeduce { using type = T; }; // exposition only
template <class T> using nodeduce_t = typename nodeduce<T>::type; // exposition only

template <class T, class Abi>
where_expression<simd_mask<T, Abi>, simd<T, Abi>> where(const typename simd<T, Abi>::mask_type&,
    simd<T, Abi>&) noexcept;

template <class T, class Abi>
const_where_expression<simd_mask<T, Abi>, simd<T, Abi>> where(const typename simd<T, Abi>::mask_type&,
    const simd<T, Abi>&) noexcept;

template <class T, class Abi>
where_expression<simd_mask<T, Abi>, simd_mask<T, Abi>> where(const nodeduce_t<simd_mask<T, Abi>>&,
    simd_mask<T, Abi>&) noexcept;

template <class T, class Abi>
const_where_expression<simd_mask<T, Abi>, simd_mask<T, Abi>> where(const nodeduce_t<simd_mask<T, Abi>>&,
    const simd_mask<T, Abi>&) noexcept;

template <class T> where_expression<bool, T> where(see below k, T& d) noexcept;
template <class T>
const_where_expression<bool, T> where(see below k, const T& d) noexcept;

// reductions [simd.reductions]
template <class T, class Abi, class BinaryOperation = plus<>>
T reduce(const simd<T, Abi>&, BinaryOperation = {});

```

```

template <class M, class V, class BinaryOperation>
typename V::value_type reduce(const const_where_expression<M, V>& x,
                             typename V::value_type identity_element, BinaryOperation binary_op);
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, plus<> binary_op = {});
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, multiplies<> binary_op);
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_and<> binary_op);
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_or<> binary_op);
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_xor<> binary_op);

template <class T, class Abi> T hmin(const simd<T, Abi>&);
template <class M, class V> typename V::value_type hmin(const const_where_expression<M, V>&);
template <class T, class Abi> T hmax(const simd<T, Abi>&);
template <class M, class V> typename V::value_type hmax(const const_where_expression<M, V>&);

// algorithms [simd.alg]
template <class T, class Abi> simd<T, Abi> min(const simd<T, Abi>&, const simd<T, Abi>&) noexcept;
template <class T, class Abi> simd<T, Abi> max(const simd<T, Abi>&, const simd<T, Abi>&) noexcept;
template <class T, class Abi>
pair<simd<T, Abi>, simd<T, Abi>> minmax(const simd<T, Abi>&, const simd<T, Abi>&) noexcept;
template <class T, class Abi>
simd<T, Abi> clamp(const simd<T, Abi>& v, const simd<T, Abi>& lo, const simd<T, Abi>& hi);
}

```

- 1 The header `<experimental/simd>` defines class templates, tag types, trait types, and function templates for element-wise operations on data-parallel objects.

(5.1.2.1) 8.2.1 simd ABI tags

[simd.abi]

```

namespace simd_abi {
    struct scalar {};
    template <int N> struct fixed_size {};
    template <typename T> inline constexpr int max_fixed_size = implementation-defined;
    template <typename T> using compatible = implementation-defined;
    template <typename T> using native = implementation-defined;
}

```

- 1 An *ABI tag* is a type in the `simd_abi` namespace that indicates a choice of size and binary representation for objects of data-parallel type. [*Note*: The intent is for the size and binary representation to depend on the target architecture. — *end note*] The ABI tag, together with a given element type implies a number of elements. ABI tag types are used as the second template argument to `simd` and `simd_mask`. [*Note*: The ABI tag is orthogonal to selecting the machine instruction set. The selected machine instruction set limits the usable ABI tag types, though (see [simd.overview]). The ABI tags enable users to safely pass objects of data-parallel type between translation unit boundaries (e.g. function calls or I/O). — *end note*]
- 2 Use of the `scalar` tag type requires data-parallel types to store a single element (i.e., `simd<T, simd_abi::scalar>::size()` returns 1). [*Note*: `scalar` is not an alias for `fixed_size<1>`. — *end note*]
- 3 The value of `max_fixed_size<T>` is at least 32.
- 4 Use of the `simd_abi::fixed_size<N>` tag type requires data-parallel types to store `N` elements (i.e. `simd<T, simd_abi::fixed_size<N>>::size()` returns `N`). `simd<T, fixed_size<N>>` and `simd_`

`mask<T, fixed_size<N>>` with $N > 0$ and $N \leq \text{max_fixed_size}<T>$ shall be supported. Additionally, for every supported `simd<T, Abi>` (see [simd.overview]), where `Abi` is an ABI tag that is not a specialization of `simd_abi::fixed_size`, $N == \text{simd}<T, Abi>::\text{size}()$ shall be supported.

[*Note:* It is unspecified whether `simd<T, fixed_size<N>>` with $N > \text{max_fixed_size}<T>$ is supported. The value of `max_fixed_size<T>` can depend on compiler flags and can change between different compiler versions. — *end note*]

[*Note:* An implementation can choose to forego ABI compatibility between differently compiled translation units for `simd` and `simd_mask` specializations using the same `simd_abi::fixed_size<N>` tag. Otherwise, the efficiency of `simd<T, Abi>` is likely to be better than for `simd<T, fixed_size<simd_size_v<T, Abi>>>` (with `Abi` not a specialization of `simd_abi::fixed_size`). — *end note*]

5 An implementation may define additional *extended ABI tag* types in the `simd_abi` namespace, to support other forms of data-parallel computation.

6 `compatible<T>` is an implementation-defined alias for an ABI tag. [*Note:* The intent is to use the ABI tag producing the most efficient data-parallel execution for the element type `T` that ensures ABI compatibility between translation units on the target architecture. — *end note*]

[*Example:* Consider a target architecture supporting the extended ABI tags `__simd128` and `__simd256`, where the `__simd256` type requires an optional ISA extension on said target architecture. Also, the target architecture does not support `long double` with either ABI tag. The implementation therefore defines

- `compatible<T>` as an alias for `__simd128` for all vectorizable `T`, except `long double`,
- and `compatible<long double>` as an alias for scalar.

— *end example*]

7 `native<T>` is an implementation-defined alias for an ABI tag. [*Note:* The intent is to use the ABI tag producing the most efficient data-parallel execution for the element type `T` that is supported on the currently targeted system. For target architectures without ISA extensions, the `native<T>` and `compatible<T>` aliases will likely be the same. For target architectures with ISA extensions, compiler flags may influence the `native<T>` alias while `compatible<T>` will be the same independent of such flags. — *end note*]

[*Example:* Consider a target architecture supporting the extended ABI tags `__simd128` and `__simd256`, where hardware support for `__simd256` only exists for floating-point types. The implementation therefore defines `native<T>` as an alias for

- `__simd256` if `T` is a floating-point type,
- and `__simd128` otherwise.

— *end example*]

```
namespace simd_abi {
    template <class T, size_t N> struct deduce { using type = see below; };
}
```

8 The member type shall be present if and only if

- `T` is a vectorizable type, and
- `simd_abi::fixed_size<N>` is supported (see [simd.abi]).

9 Where present, the member typedef type shall name an ABI tag type that satisfies

- `simd_size_v<T, type> == N`, and
- `simd<T, type>` is default constructible (see [simd.overview]).

If N is 1, the member typedef type is `simd_abi::scalar`. Otherwise, if there are multiple ABI tag types that satisfy the constraints, the member typedef type is implementation-defined. [*Note:* It is expected that extended ABI tags can produce better optimizations and thus are preferred over `simd_abi::fixed_size<N>`. — *end note*]

10 The behavior of a program that adds specializations for `deduce` is undefined.

(5.1.2.2) 8.2.2 `simd` type traits

[`simd.traits`]

```
template <class T> struct is_abi_tag { see below };
```

1 The type `is_abi_tag<T>` is a `UnaryTypeTrait` with a `BaseCharacteristic` of `true_type` if T is a standard or extended ABI tag, and `false_type` otherwise.

2 The behavior of a program that adds specializations for `is_abi_tag` is undefined.

```
template <class T> struct is_simd { see below };
```

3 The type `is_simd<T>` is a `UnaryTypeTrait` with a `BaseCharacteristic` of `true_type` if T is a specialization of the `simd` class template, and `false_type` otherwise.

4 The behavior of a program that adds specializations for `is_simd` is undefined.

```
template <class T> struct is_simd_mask { see below };
```

5 The type `is_simd_mask<T>` is a `UnaryTypeTrait` with a `BaseCharacteristic` of `true_type` if T is an specialization of the `simd_mask` class template, and `false_type` otherwise.

6 The behavior of a program that adds specializations for `is_simd_mask` is undefined.

```
template <class T> struct is_simd_flag_type { see below };
```

7 The type `is_simd_flag_type<T>` is a `UnaryTypeTrait` with a `BaseCharacteristic` of `true_type` if T is one of

- `element_aligned_tag`, or
- `vector_aligned_tag`, or
- `overaligned_tag<N>` with $N > 0$ and N an integral power of two,

and `false_type` otherwise.

8 The behavior of a program that adds specializations for `is_simd_flag_type` is undefined.

```
template <class T, class Abi = simd_abi::compatible<T>> struct simd_size { see below };
```

9 `simd_size<T, Abi>` shall have a member `value` if and only if

- T is a vectorizable type, and
- `is_abi_tag_v<Abi>` is true.

[*Note*: The rules are different from [simd.overview] — *end note*]

10 If value is present, the type `simd_size<T, Abi>` is a `BinaryTypeTrait` with a `BaseCharacteristic` of `integral_constant<size_t, N>` with `N` equal to the number of elements in a `simd<T, Abi>` object. [*Note*: If `simd<T, Abi>` is not supported for the currently targeted system, `simd_size<T, Abi>::value` produces the value `simd<T, Abi>::size()` would return if it were supported. — *end note*]

11 The behavior of a program that adds specializations for `simd_size` is undefined.

```
template <class T, class U = typename T::value_type> struct memory_alignment { see below };
```

12 `memory_alignment<T, U>` shall have a member `value` if and only if

- `is_simd_mask_v<T>` is true and `U` is `bool`, or
- `is_simd_v<T>` is true and `U` is a vectorizable type.

13 If value is present, the type `memory_alignment<T, U>` is a `BinaryTypeTrait` with a `BaseCharacteristic` of `integral_constant<size_t, N>` for some implementation-defined `N` (see [simd.copy] and [simd.mask.copy]). [*Note*: `value` identifies the alignment restrictions on pointers used for (converting) loads and stores for the given type `T` on arrays of type `U`. — *end note*]

14 The behavior of a program that adds specializations for `memory_alignment` is undefined.

(5.1.2.3) 8.2.3 Class templates `const_where_expression` and `where_expression`

[simd.whereexpr]

```
namespace std::experimental {
    inline namespace parallelism_v2 {
        template <class M, class T> class const_where_expression {
            const M mask;    // exposition only
            T& data;         // exposition only

        public:
            const_where_expression(const const_where_expression&) = delete;
            const_where_expression& operator=(const const_where_expression&) = delete;

            T operator-() const &&;
            T operator+() const &&;
            T operator~() const &&;

            template <class U, class Flags> void copy_to(U* mem, Flags f) const &&;
        };

        template <class M, class T>
        class where_expression : public const_where_expression<M, T> {
        public:
            template <class U> void operator=(U&& x) &&;
            template <class U> void operator+=(U&& x) &&;
            template <class U> void operator-=(U&& x) &&;
            template <class U> void operator*=(U&& x) &&;
            template <class U> void operator/=(U&& x) &&;
            template <class U> void operator%=(U&& x) &&;
            template <class U> void operator&=(U&& x) &&;
            template <class U> void operator|=(U&& x) &&;
            template <class U> void operator^=(U&& x) &&;
            template <class U> void operator<=(U&& x) &&;
        };
    }
}
```

```

template <class U> void operator>>=(U&& x) &&;
void operator++() &&;
void operator++(int) &&;
void operator--() &&;
void operator--(int) &&;

template <class U, class Flags> void copy_from(const U* mem, Flags) &&;
};
}

```

- 1 The class templates `const_where_expression` and `where_expression` abstract the notion of selecting elements of a given object of arithmetic or data-parallel type.
- 2 The first template argument `M` shall be cv-unqualified `bool` or a cv-unqualified `simd_mask` specialization.
- 3 If `M` is `bool`, `T` shall be a cv-unqualified arithmetic type. Otherwise, `T` shall either be `M` or `typename M::simd_type`.
- 4 In this subclause, if `M` is `bool`, `data[0]` is used interchangeably for `data`, `mask[0]` is used interchangeably for `mask`, and `M::size()` is used interchangeably for `1`.
- 5 The *selected indices* signify the integers $i \in \{j \in \mathbb{N}_0 \mid j < M::size() \wedge \text{mask}[j]\}$. The *selected elements* signify the elements `data[i]` for all selected indices `i`.
- 6 In this subclause, the type `value_type` is an alias for `T` if `M` is `bool`, or an alias for `typename T::value_type` if `is_simd_mask_v<M>` is true.
- 7 [*Note:* The `where` functions [`simd.mask.where`] initialize `mask` with the first argument to `where` and `data` with the second argument to `where`. — *end note*]

```

T operator-() const &&;
T operator+() const &&;
T operator~() const &&;

```

- 8 *Returns:* A copy of `data` with the indicated unary operator applied to all selected elements.
- 9 *Throws:* Nothing.

```

template <class U, class Flags> void copy_to(U *mem, Flags) const &&;

```

- 10 *Requires:* If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<T, U>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. If `M` is not `bool`, the largest $i \in [0, M::size())$ where `mask[i]` is true is less than the number of values pointed to by `mem`.
- 11 *Effects:* Copies the selected elements as if `mem[i] = static_cast<U>(data[i])` for all selected indices `i`.
- 12 *Throws:* Nothing.
- 13 *Remarks:* This function shall not participate in overload resolution unless
 - `is_simd_flag_type_v<Flags>` is true, and
 - either
 - `U` is `bool` and `value_type` is `bool`, or
 - `U` is a vectorizable type and `value_type` is not `bool`.

```

template <class U> void operator=(U&& x) &&;

```

14 *Effects:* Replaces `data[i]` with `static_cast<T>(std::forward<U>(x))[i]` for all selected indices `i`.

15 *Remarks:* This operator shall not participate in overload resolution unless `U` is convertible to `T`.

```
template <class U> void operator+=(U&& x) &&;
template <class U> void operator-=(U&& x) &&;
template <class U> void operator*=(U&& x) &&;
template <class U> void operator/=(U&& x) &&;
template <class U> void operator%=(U&& x) &&;
template <class U> void operator&=(U&& x) &&;
template <class U> void operator|=(U&& x) &&;
template <class U> void operator^=(U&& x) &&;
template <class U> void operator<<=(U&& x) &&;
template <class U> void operator>>=(U&& x) &&;
```

16 *Effects:* Replaces `data[i]` with `static_cast<T>(data @ std::forward<U>(x))[i]` (where `@` denotes the indicated operator) for all selected indices `i`.

17 *Remarks:* Each of these operators shall not participate in overload resolution unless the return type of `data @ std::forward<U>(x)` is convertible to `T`. It is unspecified whether the binary operator, implied by the compound assignment operator, is executed on all elements or only on the selected elements.

```
void operator++() &&;
void operator++(int) &&;
void operator--() &&;
void operator--(int) &&;
```

18 *Effects:* Applies the indicated operator to the selected elements.

19 *Remarks:* Each of these operators shall not participate in overload resolution unless the indicated operator can be applied to objects of type `T`.

```
template <class U, class Flags> void copy_from(const U *mem, Flags) &&;
```

20 *Requires:* If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<T, U>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. If `is_simd_mask_v<M>` is true, for all selected indices `i`, `i` shall be less than the number of values pointed to by `mem`.

21 *Effects:* Replaces the selected elements as if `data[i] = static_cast<value_type>(mem[i])` for all selected indices `i`.

22 *Remarks:* This function shall not participate in overload resolution unless

- `is_simd_flag_type_v<Flags>` is true, and
- either
 - `U` is `bool` and `value_type` is `bool`, or
 - `U` is a vectorizable type and `value_type` is not `bool`.

(5.1.3.1) 8.3.1 Class template simd overview

[simd.overview]

```

namespace std::experimental {
    inline namespace parallelism_v2 {
        template <class T, class Abi> class simd {
        public:
            using value_type = T;
            using reference = see below;
            using mask_type = simd_mask<T, Abi>;
            using abi_type = Abi;

            static constexpr size_t size() noexcept;

            simd() = default;

            // implicit type conversion constructor
            template <class U> simd(const simd<U, simd_abi::fixed_size<size()>>&);

            // implicit broadcast constructor (see below for constraints)
            template <class U> simd(U&& value);

            // generator constructor (see below for constraints)
            template <class G> explicit simd(G&& gen);

            // load constructor
            template <class U, class Flags> simd(const U* mem, Flags f);

            // loads [simd.load]
            template <class U, class Flags> void copy_from(const U* mem, Flags f);

            // stores [simd.store]
            template <class U, class Flags> void copy_to(U* mem, Flags f) const;

            // scalar access [simd.subscr]
            reference operator[] (size_t);
            value_type operator[] (size_t) const;

            // unary operators [simd.unary]
            simd& operator++();
            simd operator++(int);
            simd& operator--();
            simd operator--(int);
            mask_type operator!() const;
            simd operator~() const; // see below
            simd operator+() const;
            simd operator-() const;

            // binary operators [simd.binary]
            friend simd operator+ (const simd&, const simd&);
            friend simd operator- (const simd&, const simd&);
            friend simd operator* (const simd&, const simd&);
            friend simd operator/ (const simd&, const simd&);
            friend simd operator% (const simd&, const simd&);
            friend simd operator& (const simd&, const simd&);
            friend simd operator| (const simd&, const simd&);
            friend simd operator^ (const simd&, const simd&);
            friend simd operator<< (const simd&, const simd&);
            friend simd operator>> (const simd&, const simd&);

```

```

friend simd operator<<(const simd&, int);
friend simd operator>>(const simd&, int);

// compound assignment [simd.cassign]
friend simd& operator+= (simd&, const simd&);
friend simd& operator-= (simd&, const simd&);
friend simd& operator*= (simd&, const simd&);
friend simd& operator/= (simd&, const simd&);
friend simd& operator%= (simd&, const simd&);
friend simd& operator&= (simd&, const simd&);
friend simd& operator|= (simd&, const simd&);
friend simd& operator^= (simd&, const simd&);
friend simd& operator<=<=(simd&, const simd&);
friend simd& operator>=>=(simd&, const simd&);
friend simd& operator<=<=(simd&, int);
friend simd& operator>=>=(simd&, int);

// compares [simd.comparison]
friend mask_type operator==(const simd&, const simd&);
friend mask_type operator!=(const simd&, const simd&);
friend mask_type operator>=(const simd&, const simd&);
friend mask_type operator<=(const simd&, const simd&);
friend mask_type operator> (const simd&, const simd&);
friend mask_type operator< (const simd&, const simd&);
};
}
}

```

- 1 The class template `simd` is a data-parallel type. The width of a given `simd` specialization is a constant expression, determined by the template parameters.
- 2 Every specialization of `simd` shall be a complete type. The specialization `simd<T, Abi>` is supported if `T` is a vectorizable type and
 - `Abi` is `simd_abi::scalar`, or
 - `Abi` is `simd_abi::fixed_size<N>`, with `N` constrained as defined in `[simd.abi]`.

If `Abi` is an extended ABI tag, it is implementation-defined whether `simd<T, Abi>` is supported. [*Note:* The intent is for implementations to decide on the basis of the currently targeted system. — *end note*]

If `simd<T, Abi>` is not supported, the specialization shall have a deleted default constructor, deleted destructor, deleted copy constructor, and deleted copy assignment.

[*Example:* Consider an implementation that defines the extended ABI tags `__simd_x` and `__gpu_y`. When the compiler is invoked to translate to a machine that has support for the `__simd_x` ABI tag for all arithmetic types other than `long double` and no support for the `__gpu_y` ABI tag, then:

- `simd<T, simd_abi::__gpu_y>` is not supported for any `T` and has a deleted constructor.
- `simd<long double, simd_abi::__simd_x>` is not supported and has a deleted constructor.
- `simd<double, simd_abi::__simd_x>` is supported.
- `simd<long double, simd_abi::scalar>` is supported.

— *end example*]

- 3 Default initialization performs no initialization of the elements; value-initialization initializes each element with `T()`. [*Note:* Thus, default initialization leaves the elements in an indeterminate state. — *end note*]

```
static constexpr size_t size() noexcept;
```

- 4 *Returns:* The width of `simd<T, Abi>`.
- 5 Implementations should enable explicit conversion from and to implementation-defined types. This adds one or more of the following declarations to class `simd`:

```
explicit operator implementation-defined() const;
explicit simd(const implementation-defined& init);
```

[*Example:* Consider an implementation that supports the type `__vec4f` and the function `__vec4f, _vec4f_addsub(__vec4f, __vec4f)` for the currently targeted system. A user may require the use of `_vec4f_addsub` for maximum performance and thus writes:

```
using V = simd<float, simd_abi::__simd128>;
V addsub(V a, V b) {
    return static_cast<V>(_vec4f_addsub(static_cast<__vec4f>(a), static_cast<__vec4f>(b)));
}
```

— *end example*]

(5.1.3.2) 8.3.2 Element references

[`simd.reference`]

- 1 A reference is an object that refers to an element in a `simd` or `simd_mask` object. `reference::value_type` is the same type as `simd::value_type` or `simd_mask::value_type`, respectively.
- 2 Class `reference` is for exposition only. An implementation is permitted to provide equivalent functionality without providing a class with this name.

```
class reference // exposition only
{
public:
    reference() = delete;
    reference(const reference&) = delete;

    operator value_type() const noexcept;

    template <class U> reference operator=(U&& x) &&;

    template <class U> reference operator+=(U&& x) &&;
    template <class U> reference operator-=(U&& x) &&;
    template <class U> reference operator*=(U&& x) &&;
    template <class U> reference operator/=(U&& x) &&;
    template <class U> reference operator%=(U&& x) &&;
    template <class U> reference operator|=(U&& x) &&;
    template <class U> reference operator&=(U&& x) &&;
    template <class U> reference operator^=(U&& x) &&;
    template <class U> reference operator<=(U&& x) &&;
    template <class U> reference operator>=(U&& x) &&;

    reference operator++() &&;
    value_type operator++(int) &&;
    reference operator--() &&;
    value_type operator--(int) &&;

    friend void swap(reference&& a, reference&& b) noexcept;
    friend void swap(value_type& a, reference&& b) noexcept;
    friend void swap(reference&& a, value_type& b) noexcept;
};

operator value_type() const noexcept;
```

- 3 *Returns:* The value of the element referred to by `*this`.


```
template <class U> reference operator=(U&& x) &&;
```

4 *Effects:* Replaces the referred to element in `simd` or `simd_mask` with `static_cast<value_type>(std::forward<U>(x))`.

5 *Returns:* A copy of `*this`.

6 *Remarks:* This function shall not participate in overload resolution unless `declval<value_type &>() = std::forward<U>(x)` is well-formed.

```
template <class U> reference operator+=(U&& x) &&;
template <class U> reference operator-=(U&& x) &&;
template <class U> reference operator*=(U&& x) &&;
template <class U> reference operator/=(U&& x) &&;
template <class U> reference operator%=(U&& x) &&;
template <class U> reference operator|=(U&& x) &&;
template <class U> reference operator&=(U&& x) &&;
template <class U> reference operator^=(U&& x) &&;
template <class U> reference operator<=(U&& x) &&;
template <class U> reference operator>=(U&& x) &&;
```

7 *Effects:* Applies the indicated compound operator to the referred to element in `simd` or `simd_mask` and `std::forward<U>(x)`.

8 *Returns:* A copy of `*this`.

9 *Remarks:* This function shall not participate in overload resolution unless `declval<value_type &>() @= std::forward<U>(x)` (where `@=` denotes the indicated compound assignment operator) is well-formed.

```
reference operator++() &&;
reference operator--() &&;
```

10 *Effects:* Applies the indicated operator to the referred to element in `simd` or `simd_mask`.

11 *Returns:* A copy of `*this`.

12 *Remarks:* This function shall not participate in overload resolution unless the indicated operator can be applied to objects of type `value_type`.

```
value_type operator++(int) &&;
value_type operator--(int) &&;
```

13 *Effects:* Applies the indicated operator to the referred to element in `simd` or `simd_mask`.

14 *Returns:* A copy of the referred to element before applying the indicated operator.

15 *Remarks:* This function shall not participate in overload resolution unless the indicated operator can be applied to objects of type `value_type`.

```
friend void swap(reference&& a, reference&& b) noexcept;
friend void swap(value_type& a, reference&& b) noexcept;
friend void swap(reference&& a, value_type& b) noexcept;
```

16 *Effects:* Exchanges the values `a` and `b` refer to.

(5.1.3.3) 8.3.3 simd constructors

[simd.ctor]

```
template <class U> simd(U&&);
```

1 *Effects:* Constructs an object with each element initialized to the value of the argument after conversion to `value_type`.

2 *Throws:* Any exception thrown while converting the argument to `value_type`.

3 *Remarks:* Let `From` identify the type `remove_cv_t<remove_reference_t<U>>`. This constructor shall not participate in overload resolution unless:

- `From` is a vectorizable type and every possible value of `From` can be represented with type `value_type`, or
- `From` is not an arithmetic type and is implicitly convertible to `value_type`, or
- `From` is `int`, or
- `From` is `unsigned int` and `value_type` is an unsigned integral type.

```
template <class U> simd(const simd<U, simd_abi::fixed_size<size()>>& x);
```

4 *Effects:* Constructs an object where the i -th element equals `static_cast<T>(x[i])` for all $i \in [0, \text{size}())$.

5 *Remarks:* This constructor shall not participate in overload resolution unless

- `abi_type` is `simd_abi::fixed_size<size()>`, and
- every possible value of `U` can be represented with type `value_type`, and
- if both `U` and `value_type` are integral, the integer conversion rank [`conv.rank`] of `value_type` is greater than the integer conversion rank of `U`.

```
template <class G> simd(G&& gen);
```

6 *Effects:* Constructs an object where the i -th element is initialized to `gen(integral_constant<size_t, i>())`.

7 *Remarks:* This constructor shall not participate in overload resolution unless `simd(gen(integral_constant<size_t, i>()))` is well-formed for all $i \in [0, \text{size}())$.

8 The calls to `gen` are unsequenced with respect to each other. Vectorization-unsafe standard library functions may not be invoked by `gen` ([`algorithms.parallel.exec`]).

```
template <class U, class Flags> simd(const U* mem, Flags);
```

9 *Requires:* If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd, U>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. [`mem, mem + size()`] is a valid range.

10 *Effects:* Constructs an object where the i -th element is initialized to `static_cast<T>(mem[i])` for all $i \in [0, \text{size}())$.

11 *Remarks:* This constructor shall not participate in overload resolution unless

- `is_simd_flag_type_v<Flags>` is true, and
- `U` is a vectorizable type.

(5.1.3.4) 8.3.4 `simd` copy functions

[simd.copy]

```
template <class U, class Flags> void copy_from(const U* mem, Flags);
```

- 1 *Requires:* If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd, U>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. `[mem, mem + size())` is a valid range.
- 2 *Effects:* Replaces the elements of the `simd` object such that the i -th element is assigned with `static_cast<T>(mem[i])` for all $i \in [0, \text{size}())$.
- 3 *Remarks:* This function shall not participate in overload resolution unless
 - `is_simd_flag_type_v<Flags>` is true, and
 - `U` is a vectorizable type.

```
template <class U, class Flags> void copy_to(U* mem, Flags) const;
```

- 4 *Requires:* If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd, U>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. `[mem, mem + size())` is a valid range.
- 5 *Effects:* Copies all `simd` elements as if `mem[i] = static_cast<U>(operator[] (i))` for all $i \in [0, \text{size}())$.
- 6 *Remarks:* This function shall not participate in overload resolution unless
 - `is_simd_flag_type_v<Flags>` is true, and
 - `U` is a vectorizable type.

(5.1.3.5) 8.3.5 `simd` subscript operators

[simd.subscr]

```
reference operator[] (size_t i);
```

- 1 *Requires:* $i < \text{size}()$.
- 2 *Returns:* A reference (see [simd.reference]) referring to the i -th element.
- 3 *Throws:* Nothing.

```
value_type operator[] (size_t i) const;
```

- 4 *Requires:* $i < \text{size}()$.
- 5 *Returns:* The value of the i -th element.
- 6 *Throws:* Nothing.

(5.1.3.6) 8.3.6 simd unary operators

[simd.unary]

1 Effects in this subclause are applied as unary element-wise operations.

```
simd& operator++();
```

2 *Effects:* Increments every element by one.

3 *Returns:* *this.

4 *Throws:* Nothing.

```
simd operator++(int);
```

5 *Effects:* Increments every element by one.

6 *Returns:* A copy of *this before incrementing.

7 *Throws:* Nothing.

```
simd& operator--();
```

8 *Effects:* Decrements every element by one.

9 *Returns:* *this.

10 *Throws:* Nothing.

```
simd operator--(int);
```

11 *Effects:* Decrements every element by one.

12 *Returns:* A copy of *this before decrementing.

13 *Throws:* Nothing.

```
mask_type operator!() const;
```

14 *Returns:* A simd_mask object with the *i*-th element set to !operator[](*i*) for all $i \in [0, \text{size}())$.

15 *Throws:* Nothing.

```
simd operator~() const;
```

16 *Returns:* A simd object where each bit is the inverse of the corresponding bit in *this.

17 *Throws:* Nothing.

18 *Remarks:* simd::operator~() shall not participate in overload resolution unless T is an integral type.

```
simd operator+() const;
```

19 *Returns:* *this.

20 *Throws:* Nothing.

```
simd operator-() const;
```

- 21 *Returns:* A `simd` object where the i -th element is initialized to `-operator[] (i)` for all $i \in [0, \text{size}())$.
 22 *Throws:* Nothing.

(5.1.4) 8.4 `simd` non-member operations

[`simd.nonmembers`]

(5.1.4.1) 8.4.1 `simd` binary operators

[`simd.binary`]

```
friend simd operator+ (const simd& lhs, const simd& rhs);
friend simd operator- (const simd& lhs, const simd& rhs);
friend simd operator* (const simd& lhs, const simd& rhs);
friend simd operator/ (const simd& lhs, const simd& rhs);
friend simd operator% (const simd& lhs, const simd& rhs);
friend simd operator& (const simd& lhs, const simd& rhs);
friend simd operator| (const simd& lhs, const simd& rhs);
friend simd operator^ (const simd& lhs, const simd& rhs);
friend simd operator<< (const simd& lhs, const simd& rhs);
friend simd operator>> (const simd& lhs, const simd& rhs);
```

- 1 *Returns:* A `simd` object initialized with the results of the element-wise application of the indicated operator.
 2 *Throws:* Nothing.
 3 *Remarks:* Each of these operators shall not participate in overload resolution unless the indicated operator can be applied to objects of type `value_type`.

```
friend simd operator<< (const simd& v, int n);
friend simd operator>> (const simd& v, int n);
```

- 4 *Returns:* A `simd` object where the i -th element is initialized to the result of applying the indicated operator to `v[i]` and `n` for all $i \in [0, \text{size}())$.
 5 *Throws:* Nothing.
 6 *Remarks:* These operators shall not participate in overload resolution unless the indicated operator can be applied to objects of type `value_type`.

(5.1.4.2) 8.4.2 `simd` compound assignment

[`simd.cassign`]

```
friend simd& operator+= (simd& lhs, const simd& rhs);
friend simd& operator-= (simd& lhs, const simd& rhs);
friend simd& operator*= (simd& lhs, const simd& rhs);
friend simd& operator/= (simd& lhs, const simd& rhs);
friend simd& operator%= (simd& lhs, const simd& rhs);
friend simd& operator&= (simd& lhs, const simd& rhs);
friend simd& operator|= (simd& lhs, const simd& rhs);
friend simd& operator^= (simd& lhs, const simd& rhs);
friend simd& operator<<= (simd& lhs, const simd& rhs);
friend simd& operator>>= (simd& lhs, const simd& rhs);
friend simd& operator<<= (simd& lhs, int n);
friend simd& operator>>= (simd& lhs, int n);
```

- 1 *Effects:* These operators perform the indicated binary element-wise operation.
- 2 *Returns:* lhs.
- 3 *Throws:* Nothing.
- 4 *Remarks:* These operators shall not participate in overload resolution unless the indicated operator can be applied to objects of type `value_type`.

(5.1.4.3) 8.4.3 simd compare operators

[simd.comparison]

```
friend mask_type operator==(const simd&, const simd&);
friend mask_type operator!=(const simd&, const simd&);
friend mask_type operator>=(const simd&, const simd&);
friend mask_type operator<=(const simd&, const simd&);
friend mask_type operator> (const simd&, const simd&);
friend mask_type operator< (const simd&, const simd&);
```

- 1 *Returns:* A `simd_mask` object initialized with the results of the element-wise application of the indicated operator.
- 2 *Throws:* Nothing.

(5.1.4.4) 8.4.4 simd reductions

[simd.reductions]

- 1 In this subclause, `BinaryOperation` shall be a binary element-wise operation.

```
template <class T, class Abi, class BinaryOperation = plus<>>
T reduce(const simd<T, Abi>& x, BinaryOperation binary_op = {});
```

- 2 *Requires:* `binary_op` shall be callable with two arguments of type `T` returning `T`, or callable with two arguments of type `simd<T, A1>` returning `simd<T, A1>` for every `A1` that is an ABI tag type.
- 3 *Returns:* `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in [0, \text{size}())$.
- 4 *Throws:* Any exception thrown from `binary_op`.
- 5 [*Note:* This overload of `reduce` does not require an initial value because `x` is guaranteed to be non-empty. — end note]

```
template <class M, class V, class BinaryOperation>
typename V::value_type reduce(const const_where_expression<M, V>& x, typename V::value_type identity_element,
                             BinaryOperation binary_op);
```

- 6 *Requires:* `binary_op` shall be callable with two arguments of type `T` returning `T`, or callable with two arguments of type `simd<T, A1>` returning `simd<T, A1>` for every `A1` that is an ABI tag type. The results of `binary_op(identity_element, x)` and `binary_op(x, identity_element)` shall be equal to `x` for all finite values `x` representable by `V::value_type`.
- 7 *Returns:* If `none_of(x.mask)`, returns `identity_element`. Otherwise, returns `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.
- 8 *Throws:* Any exception thrown from `binary_op`.

```
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, plus<> binary_op = {});
```

9 *Returns:* If `none_of(x.mask)`, returns 0. Otherwise, returns `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.

10 *Throws:* Nothing.

```
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, multiplies<> binary_op);
```

11 *Returns:* If `none_of(x.mask)`, returns 1. Otherwise, returns `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.

12 *Throws:* Nothing.

```
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_and<> binary_op);
```

13 *Requires:* `is_integral_v<V::value_type>` is true.

14 *Returns:* If `none_of(x.mask)`, returns `~V::value_type()`. Otherwise, returns `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.

15 *Throws:* Nothing.

```
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_or<> binary_op);
template <class M, class V>
typename V::value_type reduce(const const_where_expression<M, V>& x, bit_xor<> binary_op);
```

16 *Requires:* `is_integral_v<V::value_type>` is true.

17 *Returns:* If `none_of(x.mask)`, returns 0. Otherwise, returns `GENERALIZED_SUM(binary_op, x.data[i], ...)` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.

18 *Throws:* Nothing.

```
template <class T, class Abi> T hmin(const simd<T, Abi>& x);
```

19 *Returns:* The value of an element `x[j]` for which `x[j] <= x[i]` for all $i \in [0, \text{size}())$.

20 *Throws:* Nothing.

```
template <class M, class V> typename V::value_type hmin(const const_where_expression<M, V>& x);
```

21 *Returns:* If `none_of(x.mask)`, the return value is `numeric_limits<V::value_type>::max()`. Otherwise, returns the value of an element `x.data[j]` for which `x.mask[j] == true` and `x.data[j] <= x.data[i]` for all $i \in \{j \in \mathbb{N}_0 \mid j < \text{size}() \wedge x.\text{mask}[j]\}$.

22 *Throws:* Nothing.

```
template <class T, class Abi> T hmax(const simd<T, Abi>& x);
```

23 *Returns:* The value of an element `x[j]` for which `x[j] >= x[i]` for all $i \in [0, \text{size}())$.

24 *Throws:* Nothing.

```
template <class M, class V> typename V::value_type hmax(const const_where_expression<M, V>& x);
```

25 *Returns:* If `none_of(x.mask)`, the return value is `numeric_limits<V::value_type>::lowest()`.
 Otherwise, returns the value of an element `x.data[j]` for which `x.mask[j] == true` and `x.data[j] >= x.data[i]` for all $i \in \{j \in \mathbb{N}_0 | j < \text{size}() \wedge x.\text{mask}[j]\}$.

26 *Throws:* Nothing.

(5.1.4.5) 8.4.5 simd casts

[simd.casts]

```
template <class T, class U, class Abi> see below simd_cast(const simd<U, Abi>& x);
```

- 1 Let `To` identify `T::value_type` if `is_simd_v<T>` is true, or `T` otherwise.
- 2 *Returns:* A `simd` object with the i -th element initialized to `static_cast<To>(x[i])` for all $i \in [0, \text{size}())$.
- 3 *Throws:* Nothing.
- 4 *Remarks:* The function shall not participate in overload resolution unless
- every possible value of type `U` can be represented with type `To`, and
 - either
 - `is_simd_v<T>` is false, or
 - `T::size() == simd<U, Abi>::size()` is true.
- 5 The return type is
- `T` if `is_simd_v<T>` is true, otherwise
 - `simd<T, Abi>` if `U` is `T`, otherwise
 - `simd<T, simd_abi::fixed_size<simd<U, Abi>::size()>>`.

```
template <class T, class U, class Abi> see below static_simd_cast(const simd<U, Abi>& x);
```

- 6 Let `To` identify `T::value_type` if `is_simd_v<T>` is true or `T` otherwise.
- 7 *Returns:* A `simd` object with the i -th element initialized to `static_cast<To>(x[i])` for all $i \in [0, \text{size}())$.
- 8 *Throws:* Nothing.
- 9 *Remarks:* The function shall not participate in overload resolution unless either
- `is_simd_v<T>` is false, or
 - `T::size() == simd<U, Abi>::size()` is true.
- 10 The return type is
- `T` if `is_simd_v<T>` is true, otherwise
 - `simd<T, Abi>` if either `U` is `T` or `U` and `T` are integral types that only differ in signedness, otherwise
 - `simd<T, simd_abi::fixed_size<simd<U, Abi>::size()>>`.

```
template <class T, class Abi>
fixed_size_simd<T, simd_size_v<T, Abi>> to_fixed_size(const simd<T, Abi>& x) noexcept;
template <class T, class Abi>
fixed_size_simd_mask<T, simd_size_v<T, Abi>> to_fixed_size(const simd_mask<T, Abi>& x) noexcept;
```


11 *Returns:* A data-parallel object with the i -th element initialized to $x[i]$ for all $i \in [0, \text{size}())$.

```
template <class T, int N> native_simd<T> to_native(const fixed_size_simd<T, N>& x) noexcept;
template <class T, int N> native_simd_mask<T> to_native(const fixed_size_simd_mask<T, N>& x) noexcept;
```

12 *Returns:* A data-parallel object with the i -th element initialized to $x[i]$ for all $i \in [0, \text{size}())$.

13 *Remarks:* These functions shall not participate in overload resolution unless `simd_size_v<T, simd_abi::native<T>> == N` is true.

```
template <class T, int N> simd<T> to_compatible(const fixed_size_simd<T, N>& x) noexcept;
template <class T, int N> simd_mask<T> to_compatible(const fixed_size_simd_mask<T, N>& x) noexcept;
```

14 *Returns:* A data-parallel object with the i -th element initialized to $x[i]$ for all $i \in [0, \text{size}())$.

15 *Remarks:* These functions shall not participate in overload resolution unless `simd_size_v<T, simd_abi::compatible<T>> == N` is true.

```
template <size_t... Sizes, class T, class Abi>
tuple<simd<T, simd_abi::deduce_t<T, Sizes>>...> split(const simd<T, Abi>& x);
template <size_t... Sizes, class T, class Abi>
tuple<simd_mask<T, simd_abi::deduce_t<T, Sizes>>...> split(const simd_mask<T, Abi>& x);
```

16 *Returns:* A tuple of data-parallel objects with the i -th `simd/simd_mask` element of the j -th tuple element initialized to the value of the element in x with index $i + \text{sum of the first } j \text{ values in the Sizes pack}$.

17 *Remarks:* These functions shall not participate in overload resolution unless the sum of all values in the `Sizes` pack is equal to `simd_size_v<T, Abi>`.

```
template <class V, class Abi>
array<V, simd_size_v<typename V::value_type, Abi> / V::size()> split(
    const simd<typename V::value_type, Abi>& x);
template <class V, class Abi>
array<V, simd_size_v<typename V::value_type, Abi> / V::size()> split(
    const simd_mask<typename V::value_type, Abi>& x);
```

18 *Returns:* An array of data-parallel objects with the i -th `simd/simd_mask` element of the j -th array element initialized to the value of the element in x with index $i + j * V::size()$.

19 *Remarks:* These functions shall not participate in overload resolution unless

- `simd_size_v<typename V::value_type, Abi>` is an integral multiple of `V::size()`, and
- for the overload with a `simd` parameter `is_simd_v<V>` is true, for the overload with a `simd_mask` parameter `is_simd_mask_v<V>` is true.

```
template <class T, class... Abis>
simd<T, simd_abi::deduce_t<T, (simd_size_v<T, Abis> + ...)>> concat(const simd<T, Abis>&... xs);
template <class T, class... Abis>
simd_mask<T, simd_abi::deduce_t<T, (simd_size_v<T, Abis> + ...)>> concat(const simd_mask<T, Abis>&... xs);
```

20 *Returns:* A data-parallel object initialized with the concatenated values in the `xs` pack of data-parallel objects: The i -th `simd/simd_mask` element of the j -th parameter in the `xs` pack is copied to the return value's element with index $i + \text{the sum of the width of the first } j \text{ parameters in the } xs \text{ pack}$.

(5.1.4.6) 8.4.6 simd algorithms

[simd.alg]

```
template <class T, class Abi> simd<T, Abi> min(const simd<T, Abi>& a, const simd<T, Abi>& b) noexcept;
```

1 *Returns:* The result of element-wise application of `std::min(a[i], b[i])` for all $i \in [0, \text{size}())$.

```
template <class T, class Abi> simd<T, Abi> max(const simd<T, Abi>& a, const simd<T, Abi>& b) noexcept;
```

2 *Returns:* The result of element-wise application of `std::max(a[i], b[i])` for all $i \in [0, \text{size}())$.

```
template <class T, class Abi>
pair<simd<T, Abi>, simd<T, Abi>> minmax(const simd<T, Abi>& a, const simd<T, Abi>& b) noexcept;
```

3 *Returns:* A pair initialized with

- the result of element-wise application of `std::min(a[i], b[i])` for all $i \in [0, \text{size}())$ in the first member, and
- the result of element-wise application of `std::max(a[i], b[i])` for all $i \in [0, \text{size}())$ in the second member.

```
template <class T, class Abi>
simd<T, Abi> clamp(const simd<T, Abi>& v, const simd<T, Abi>& lo, const simd<T, Abi>& hi);
```

4 *Requires:* No element in `lo` shall be greater than the corresponding element in `hi`.

5 *Returns:* The result of element-wise application of `std::clamp(v[i], lo[i], hi[i])` for all $i \in [0, \text{size}())$.

(5.1.4.7) 8.4.7 simd math library

[simd.math]

1 For each set of overloaded functions within `<cmath>`, there shall be additional overloads sufficient to ensure that if any argument corresponding to a double parameter has type `simd<T, Abi>`, where `is_floating_point_v<T>` is `true`, then:

- All arguments corresponding to double parameters shall be convertible to `simd<T, Abi>`.
- All arguments corresponding to `double*` parameters shall be of type `simd<T, Abi>*`.
- All arguments corresponding to parameters of integral type `U` shall be convertible to `fixed_size_simd<U, simd_size_v<T, Abi>>`.
- All arguments corresponding to `U*`, where `U` is integral, shall be of type `fixed_size_simd<U, simd_size_v<T, Abi>>*`.
- If the corresponding return type is `double`, the return type of the additional overload is `simd<T, Abi>`. Otherwise, if the corresponding return type is `bool`, the return type of the additional overload is `simd_mask<T, Abi>`. Otherwise, the return type is `fixed_size_simd<R, simd_size_v<T, Abi>>`, with `R` denoting the corresponding return type.

It is unspecified whether a call to these overloads with arguments that are all convertible to `simd<T, Abi>` but are not of type `simd<T, Abi>` is well-formed.

- 2 Each function overload produced by the above rules applies the indicated `<cmath>` function element-wise. The results per element are not required to be bitwise equal to the application of the function which is overloaded for the element type.
- 3 The behavior is undefined if a domain, pole, or range error occurs when the input argument(s) are applied to the indicated `<cmath>` function.
- 4 If `abs` is called with an argument of type `simd<X, Abi>` for which `is_unsigned_v<X>` is true, the program is ill-formed.

(5.1.5) 8.5 Class template `simd_mask` [simd.mask.class]

(5.1.5.1) 8.5.1 Class template `simd_mask` overview [simd.mask.overview]

```
namespace std::experimental {
    inline namespace parallelism_v2 {
        template <class T, class Abi> class simd_mask {
        public:
            using value_type = bool;
            using reference = see below;
            using simd_type = simd<T, Abi>;
            using abi_type = Abi;

            static constexpr size_t size() noexcept;

            simd_mask() = default;

            // broadcast constructor
            explicit simd_mask(value_type) noexcept;

            // implicit type conversion constructor
            template <class U> simd_mask(const simd_mask<U, simd_abi::fixed_size<size()>>&) noexcept;

            // load constructor
            template <class Flags> simd_mask(const value_type* mem, Flags);

            // loads [simd.mask.copy]
            template <class Flags> void copy_from(const value_type* mem, Flags);
            template <class Flags> void copy_to(value_type* mem, Flags) const;

            // scalar access [simd.mask.subscr]
            reference operator[] (size_t);
            value_type operator[] (size_t) const;

            // unary operators [simd.mask.unary]
            simd_mask operator!() const noexcept;

            // simd_mask binary operators [simd.mask.binary]
            friend simd_mask operator&& (const simd_mask&, const simd_mask&) noexcept;
            friend simd_mask operator|| (const simd_mask&, const simd_mask&) noexcept;
            friend simd_mask operator& (const simd_mask&, const simd_mask&) noexcept;
            friend simd_mask operator| (const simd_mask&, const simd_mask&) noexcept;
            friend simd_mask operator^ (const simd_mask&, const simd_mask&) noexcept;
```

```

// simd_mask compound assignment [simd.mask.cassign]
friend simd_mask& operator&=(simd_mask&, const simd_mask&) noexcept;
friend simd_mask& operator|=(simd_mask&, const simd_mask&) noexcept;
friend simd_mask& operator^=(simd_mask&, const simd_mask&) noexcept;

// simd_mask compares [simd.mask.comparison]
friend simd_mask operator==(const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator!=(const simd_mask&, const simd_mask&) noexcept;
};
}

```

- 1 The class template `simd_mask` is a data-parallel type with the element type `bool`. The width of a given `simd_mask` specialization is a constant expression, determined by the template parameters. Specifically, `simd_mask<T, Abi>::size() == simd<T, Abi>::size()`.
- 2 Every specialization of `simd_mask` shall be a complete type. The specialization `simd_mask<T, Abi>` is supported if `T` is a vectorizable type and
 - `Abi` is `simd_abi::scalar`, or
 - `Abi` is `simd_abi::fixed_size<N>`, with `N` constrained as defined in `[simd.abi]`.

If `Abi` is an extended ABI tag, it is implementation-defined whether `simd_mask<T, Abi>` is supported. [*Note:* The intent is for implementations to decide on the basis of the currently targeted system. — *end note*]

If `simd_mask<T, Abi>` is not supported, the specialization shall have a deleted default constructor, deleted destructor, deleted copy constructor, and deleted copy assignment.

- 3 Default initialization performs no initialization of the elements; value-initialization initializes each element with `false`. [*Note:* Thus, default initialization leaves the elements in an indeterminate state. — *end note*]

```
static constexpr size_t size() noexcept;
```

- 4 *Returns:* The width of `simd<T, Abi>`.
- 5 Implementations should enable explicit conversion from and to implementation-defined types. This adds one or more of the following declarations to class `simd_mask`:

```
explicit operator implementation-defined() const;
explicit simd_mask(const implementation-defined& init);
```

- 6 The member type `reference` has the same interface as `simd<T, Abi>::reference`, except its `value_type` is `bool`. (`[simd.reference]`)

(5.1.5.2) 8.5.2 `simd_mask` constructors [simd.mask.ctor]

```
explicit simd_mask(value_type x) noexcept;
```

- 1 *Effects:* Constructs an object with each element initialized to `x`.

```
template <class U> simd_mask(const simd_mask<U, simd_abi::fixed_size<size()>>& x) noexcept;
```

- 2 *Effects:* Constructs an object of type `simd_mask` where the i -th element equals `x[i]` for all $i \in [0, \text{size}())$.
- 3 *Remarks:* This constructor shall not participate in overload resolution unless `abi_type` is `simd_abi::fixed_size<size()>`.

```
template <class Flags> simd_mask(const value_type* mem, Flags);
```

- 4 **Requires:** If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd_mask>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. `[mem, mem + size())` is a valid range.
- 5 **Effects:** Constructs an object where the i -th element is initialized to `mem[i]` for all $i \in [0, \text{size}())$.
- 6 **Remarks:** This constructor shall not participate in overload resolution unless `is_simd_flag_type_v<Flags>` is true.

(5.1.5.3) 8.5.3 `simd_mask` copy functions

[simd.mask.copy]

```
template <class Flags> void copy_from(const value_type* mem, Flags);
```

- 1 **Requires:** If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd_mask>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. `[mem, mem + size())` is a valid range.
- 2 **Effects:** Replaces the elements of the `simd_mask` object such that the i -th element is replaced with `mem[i]` for all $i \in [0, \text{size}())$.
- 3 **Remarks:** This function shall not participate in overload resolution unless `is_simd_flag_type_v<Flags>` is true.

```
template <class Flags> void copy_to(value_type* mem, Flags) const;
```

- 4 **Requires:** If the template parameter `Flags` is `vector_aligned_tag`, `mem` shall point to storage aligned by `memory_alignment_v<simd_mask>`. If the template parameter `Flags` is `overaligned_tag<N>`, `mem` shall point to storage aligned by `N`. If the template parameter `Flags` is `element_aligned_tag`, `mem` shall point to storage aligned by `alignof(U)`. `[mem, mem + size())` is a valid range.
- 5 **Effects:** Copies all `simd_mask` elements as if `mem[i] = operator[] (i)` for all $i \in [0, \text{size}())$.
- 6 **Remarks:** This function shall not participate in overload resolution unless `is_simd_flag_type_v<Flags>` is true.

(5.1.5.4) 8.5.4 `simd_mask` subscript operators

[simd.mask.subscr]

```
reference operator[] (size_t i);
```

- 1 **Requires:** $i < \text{size}()$.
- 2 **Returns:** A reference (see [simd.reference]) referring to the i -th element.
- 3 **Throws:** Nothing.

```
value_type operator[] (size_t i) const;
```

- 4 *Requires:* $i < \text{size}()$.
- 5 *Returns:* The value of the i -th element.
- 6 *Throws:* Nothing.

(5.1.5.5) 8.5.5 `simd_mask` unary operators [simd.mask.unary]

```
simd_mask operator!() const noexcept;
```

- 1 *Returns:* The result of element-wise application of `operator!`.

(5.1.6) 8.6 `simd_mask` non-member operations [simd.mask.nonmembers]

(5.1.6.1) 8.6.1 `simd_mask` binary operators [simd.mask.binary]

```
friend simd_mask operator&&(const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator|| (const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator& (const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator| (const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator^ (const simd_mask&, const simd_mask&) noexcept;
```

- 1 *Returns:* A `simd_mask` object initialized with the results of the element-wise application of the indicated operator.

(5.1.6.2) 8.6.2 `simd_mask` compound assignment [simd.mask.cassign]

```
friend simd_mask& operator&=(simd_mask& lhs, const simd_mask& rhs) noexcept;
friend simd_mask& operator|=(simd_mask& lhs, const simd_mask& rhs) noexcept;
friend simd_mask& operator^=(simd_mask& lhs, const simd_mask& rhs) noexcept;
```

- 1 *Effects:* These operators perform the indicated binary element-wise operation.
- 2 *Returns:* `lhs`.

(5.1.6.3) 8.6.3 `simd_mask` comparisons [simd.mask.comparison]

```
friend simd_mask operator==(const simd_mask&, const simd_mask&) noexcept;
friend simd_mask operator!=(const simd_mask&, const simd_mask&) noexcept;
```

- 1 *Returns:* An object initialized with the results of the element-wise application of the indicated operator.

(5.1.6.4) 8.6.4 `simd_mask` reductions [simd.mask.reductions]

```
template <class T, class Abi> bool all_of(const simd_mask<T, Abi>& k) noexcept;
```

- 1 *Returns:* true if all boolean elements in `k` are true, false otherwise.

```
template <class T, class Abi> bool any_of(const simd_mask<T, Abi>& k) noexcept;
```

2 *Returns:* true if at least one boolean element in *k* is true, false otherwise.

```
template <class T, class Abi> bool none_of(const simd_mask<T, Abi>& k) noexcept;
```

3 *Returns:* true if none of the boolean elements in *k* is true, false otherwise.

```
template <class T, class Abi> bool some_of(const simd_mask<T, Abi>& k) noexcept;
```

4 *Returns:* true if at least one of the boolean elements in *k* is true and at least one of the boolean elements *k* is false, false otherwise.

```
template <class T, class Abi> int popcount(const simd_mask<T, Abi>& k) noexcept;
```

5 *Returns:* The number of boolean elements in *k* that are true.

```
template <class T, class Abi> int find_first_set(const simd_mask<T, Abi>& k);
```

6 *Requires:* *any_of(k)* returns true.

7 *Returns:* The lowest element index *i* where *k[i]* is true.

```
template <class T, class Abi> int find_last_set(const simd_mask<T, Abi>& k);
```

8 *Requires:* *any_of(k)* returns true.

9 *Returns:* The greatest element index *i* where *k[i]* is true.

```
bool all_of(see below) noexcept;
bool any_of(see below) noexcept;
bool none_of(see below) noexcept;
bool some_of(see below) noexcept;
int popcount(see below) noexcept;
```

10 *Returns:* *all_of* and *any_of* return their arguments; *none_of* returns the negation of its argument; *some_of* returns false; *popcount* returns the integral representation of its argument; *find_first_set* and *find_last_set* return 0.

11 *Remarks:* The functions shall not participate in overload resolution unless the argument is of type *bool*.

```
int find_first_set(see below) noexcept;
int find_last_set(see below) noexcept;
```

12 *Requires:* The value of the argument is true.

13 *Returns:* 0.

14 *Remarks:* The functions shall not participate in overload resolution unless the argument is of type *bool*.

```

template <class T, class Abi>
where_expression<simd_mask<T, Abi>, simd<T, Abi>> where(const typename simd<T, Abi>::mask_type& k,
                                                         simd<T, Abi>& v) noexcept;

template <class T, class Abi>
const_where_expression<simd_mask<T, Abi>, simd<T, Abi>> where(
    const typename simd<T, Abi>::mask_type& k, const simd<T, Abi>& v) noexcept;

template <class T, class Abi>
where_expression<simd_mask<T, Abi>, simd_mask<T, Abi>> where(const nodeduce_t<simd_mask<T, Abi>>& k,
                                                             simd_mask<T, Abi>& v) noexcept;

template <class T, class Abi>
const_where_expression<simd_mask<T, Abi>, simd_mask<T, Abi>> where(
    const nodeduce_t<simd_mask<T, Abi>>& k, const simd_mask<T, Abi>& v) noexcept;

```

- 1 *Returns:* An object [simd.whereexpr] with mask and data initialized with k and v respectively.

```

template <class T> where_expression<bool, T> where(see below k, T& v) noexcept;
template <class T>
const_where_expression<bool, T> where(see below k, const T& v) noexcept;

```

- 2 *Remarks:* The functions shall not participate in overload resolution unless
- T is neither a simd nor a simd_mask specialization, and
 - the first argument is of type bool.
- 3 *Returns:* An object [simd.whereexpr] with mask and data initialized with k and v respectively.

6

FEATURE DETECTION MACROS

For the purposes of SD-6, feature detection initially will be provided through the shipping vehicle (TS) itself. For a standalone feature detection macro, I recommend `__cpp_lib_simd`.

7

DISCUSSION

7.1

MEMBER TYPES

The member types may not seem obvious. Rationales:

`value_type`

In the spirit of the `value_type` member of STL containers, this type denotes the *logical* type of the values in the vector.

`reference`

Used as the return type of the non-const scalar subscript operator.

`mask_type`

The natural `simd_mask` type for this `simd` instantiation. This type is used as return type of compares and write-mask on assignments.

`simd_type`

The natural `simd` type for this `simd_mask` instantiation.

`size_type`

Standard member type used for `size()` and `operator[]`.

`abi_type`

The `Abi` template parameter to `simd`.

7.2

CONVERSIONS

The `simd` conversion constructor only allows implicit conversion from `simd` template instantiations with the same `Abi` type and compatible `value_type`. Discussion in SG1 showed clear preference for only allowing implicit conversion between integral types that only differ in signedness. All other conversions could be implemented via an explicit conversion constructor. The alternative (preferred) is to use `simd_cast` consistently for all other conversions.

After more discussion on the LEWG reflector, in Issaquah, and between me and Jens, we modified conversions to be even more conservative. No implicit conversion will ever allow a narrowing conversion of the element type (and signed - unsigned is narrowing in both directions). Also, implicit conversions are only enabled between `fixed_size` instances. All other ABI tags require explicit conversions in every case. This is very conservative and I expect TS feedback on this issue to ask for more (non-narrowing) implicit conversions.

7.3

BROADCAST CONSTRUCTOR

The `simd` broadcast constructor is not declared `explicit` to ease the use of scalar prvalues in expressions involving data-parallel operations. The operations where such a conversion should not be implicit consequently need to use SFINAE / concepts to inhibit the conversion.

Experience from Vc shows that the situation is different for `simd_mask`, where an implicit conversion from `bool` typically hides an error. (Since there is little use for broadcasting `true` or `false`.)

7.4

ALIASING OF SUBSCRIPT OPERATORS

The subscript operators return an rvalue. The `const` overload returns a copy of the element. The non-`const` overload returns a smart reference. This reference behaves mostly like an lvalue reference, but without the requirement to implement assignment via type punning. At this point the specification of the smart reference is very conservative / restrictive: The reference type is neither copyable nor movable. The intention is to avoid users to program like the operator returned an lvalue reference. The return type is significantly larger than an lvalue reference and harder to optimize when passed around. The restriction thus forces users to do element modification directly on the `simd/ simd_mask` objects.

Guidance from SG1 at JAX 2016:

Poll: Should subscript operator return an lvalue reference?

SF	F	N	A	SA
0	6	10	2	1

Poll: Should subscript operator return a “smart reference”?

SF	F	N	A	SA
1	7	10	0	0

7.5

COMPOUND ASSIGNMENT

The semantics of compound assignment would allow less strict implicit conversion rules. Consider `simd<int>() *= double()`: the corresponding binary multiplication operator would not compile because the implicit conversion to `simd<double>` is non-portable. Compound assignment, on the other hand, implies an implicit conversion back to the type of the expression on the left of the assignment operator. Thus, it is possible to define compound operators that execute the operation correctly on the promoted type without sacrificing portability. There are two arguments for not relaxing the rules for compound assignment, though:

1. Consistency: The conversion of an expression with compound assignment to a binary operator might make it ill-formed.
2. The implicit conversion in the `int * double` case could be expensive and unintended. This is already a problem for builtin types, where many developers multiply `float` variables with `double` prvalues, though.

7.6

RETURN TYPE OF MASKED ASSIGNMENT OPERATORS

The assignment operators of the type returned by `where(mask, simd)` could return one of:

- A reference to the `simd` object that was modified.
- A temporary `simd` object that only contains the elements where the `simd_mask` is `true`.
- A reference to the `where_expression` object.
- Nothing (i.e. `void`).

My first choice was a reference to the modified `simd` object. However, then the statement `(where(x < 0, x) *= -1) += 2` may be surprising: it adds 2 to all vector entries, independent of the mask. Likewise, `y += (where(x < 0, x) *= -1)` has a possibly confusing interpretation because of the `simd_mask` in the middle of the expression.

Consider that write-masked assignment is used as a replacement for `if`-statements. Using `void` as return type therefore is a more fitting choice because `if`-statements have no return value. By declaring the return type as `void` the above expressions become ill-formed, which seems to be the best solution for guiding users to write maintainable code and express intent clearly.

7.7

FUNDAMENTAL SIMD TYPE OR NOT?

7.7.1

THE ISSUE

There was substantial discussion on the reflectors and SG1 meetings over the question whether C++ should define a fundamental, native SIMD type (let us call it `fundamental<T>`) and additionally a generic data-parallel type which supports an arbitrary number of elements (call it `arbitrary<T, N>`). The alternative to defining both types is to only define `arbitrary<T, N = default_size<T>>`, since it encompasses the `fundamental<T>` type.

With regard to this proposal this second approach would add a third template parameter to `simd` and `simd_mask` as shown in Listing 1.

7.7.2

STANDPOINTS

The controversy is about how the flexibility of a type with arbitrary `N` is presented to the users. Is there a (clear) distinction between a “fundamental” type with target-dependent (i.e. fixed) `N` and a higher-level abstraction with arbitrary `N` which can

```

1 template <class T, size_t N = simd_size_v<T, simd_abi::compatible<T>>,
2         class Abi = simd_abi::compatible<T>>
3 class simd;

```

Listing 1: Possible declaration of the class template parameters of a `simd` class with arbitrary width.

potentially compile to inefficient machine code? Or should the C++ standard only define `arbitrary` and set it to a default `N` value that corresponds to the target-dependent `N`. Thus, the default `N`, of `arbitrary` would correspond to `fundamental`.

It is interesting to note that `arbitrary<T, 1>` is the class variant of `T`. Consequently, if we say there is no need for a `fundamental` type then we could argue for the deprecation of the builtin arithmetic types, in favor of `arbitrary<T, 1>`. [*Note:* This is an academic discussion, of course. — *end note*]

The author has implemented a library where a clear distinction is made between `fundamental<T, Abi>` and `arbitrary<T, N>`. The documentation and all teaching material says that the user should program with `fundamental`. The `arbitrary` type should be used in special circumstances, or wherever `fundamental` works with the `arbitrary` type in its interfaces (e.g. for `gather & scatter` or the `ldexp & frexp` functions).

7.7.3

ISSUES

The definition of two separate class templates can alleviate some source compatibility issues resulting from different `N` on different target systems. Consider the simplest example of a multiplication of an `int` vector with a `float` vector:

```

arbitrary<float>() * arbitrary<int>(); // compiles for some targets, fails for others
fundamental<float>() * fundamental<int>(); // never compiles, requires explicit cast

```

The `simd<T>` operators are specified in such a way that source compatibility is ensured. For a type with user definable `N`, the binary operators should work slightly different with regard to implicit conversions. Most importantly, `arbitrary<T, N>` solves the issue of portable code containing mixed integral and floating-point values. A user would typically create aliases such as:

```

using floatvec = simd<float>;
using intvec = arbitrary<int, floatvec::size()>;
using doublevec = arbitrary<int, floatvec::size()>;

```

Objects of types `floatvec`, `intvec`, and `doublevec` will work together, independent of the target system.

Obviously, these type aliases are basically the same if the `N` parameter of `arbitrary` has a default value:

```
using floatvec = arbitrary<float>;
using intvec = arbitrary<int, floatvec::size()>;
using doublevec = arbitrary<int, floatvec::size()>;
```

The ability to create these aliases is not the issue. Seeing the need for using such a pattern is the issue. Typically, a developer will think no more of it if his code compiles on his machine. If `arbitrary<float>() * arbitrary<int>()` just happens to compile (which is likely), then this is the code that will get checked in to the repository. Note that with the existence of the `fundamental` class template, the `N` parameter of the `arbitrary` class would not have a default value and thus force the user to think a second longer about portability.

7.7.4

PROGRESS

SG1 Guidance at JAX 2016:

Poll: Specify `simd` width using ABI tag, with a special template tag for fixed size.

SF	F	N	A	SA
3	7	0	0	1

Poll: Specify `simd` width using `<T, N, abi>`, where `abi` is not specified by the user.

SF	F	N	A	SA
1	2	5	2	1

At the Jacksonville meeting, SG1 decided to continue with the `simd<T, Abi>` class template, with the addition of a new `Abi` type that denotes a user-requested number of elements in the vector (`simd_abi::fixed_size<N>`). This has the following implications:

- There is only one class template with a common interface for *fundamental* and *arbitrary* (`fixed_size`) vector types.
- There are slight differences in the conversion semantics for `simd` types with the `fixed_size` `Abi` type. This may look like the `vector<bool>` mistake all over again. I'll argue below why I believe this is not the case.
- The *fundamental* class instances could be implemented in such a way that they do not guarantee ABI compatibility on a given architecture where translation units are compiled with different compiler flags (for micro-architectural differences).
- The `fixed_size` class instances, on the other hand, could be implemented to be the ABI stable types (if an implementation thinks this is an important feature).

In implementation terms this means that *fundamental* types are allowed to be passed via registers on function calls. `fixed_size` types can be implemented in such a way that they are only passed via the stack, and thus an implementation only needs to ensure equal alignment and memory representation across TU borders for a given `T`, `N`.

The conversion differences between the *fundamental* and `fixed_size` class template instances are the main motivation for having a distinction (cf. discussion above). The differences are chosen such that, in general, *fundamental* types are more restrictive and do not turn into `fixed_size` types on any operation that involves no `fixed_size` types. Operations of `fixed_size` types allow easier use of mixed precision code as long as no elements need to be dropped / generated (i.e. the number of elements of all involved `simd` objects is equal or a builtin arithmetic type is broadcast).

Examples:

1. Mixed int-float operations

```
1 using floatv = simd<float>; // native ABI
2 using float_sized_abi = simd_abi::fixed_size<floatv::size()>;
3 using intv = simd<int, float_sized_abi>;
4
5 auto x = floatv() + intv(); /*! \label{lstline:1} */
6 intv y = floatv() + intv(); /*! \label{lstline:2} */
```

Line ?? is well-formed: It states that N ($= \text{floatv}::\text{size}()$) additions shall be executed concurrently. The type of `x` is `simd<float>`, because it stores N elements and both types `intv` and `floatv` are implicitly convertible to `simd<float>`. Line ?? is also well-formed because implicit conversion from `simd<T, Abi>` to `simd<U, simd_abi::fixed_size<N>>` is allowed whenever $N == \text{simd}<T, Abi>::\text{size}()$.

2. Native int vectors

```
1 using intv = simd<int>; // native ABI
2 using int_sized_abi = simd_abi::fixed_size<intv::size()>;
3 using floatv = simd<float, int_sized_abi>;
4
5 auto x = floatv() + intv(); /*! \label{lstline:3} */
6 intv y = floatv() + intv(); /*! \label{lstline:4} */
```

Line ?? is well-formed: It states that N ($= \text{intv}::\text{size}()$) additions shall be executed concurrently. The type of `x` is `simd<simd<float>, int_sized_abi>` (i.e. `floatv`) and never `simd<float>`, because ...

... the `Abi` types of `intv` and `floatv` are not equal.

... either `simd<float>::size() != N` or `intv` is not implicitly convertible to `simd<float>`.

... the last rule for `commonabi(V0, V1, T)` sets the `Abi` type to `int_sized_abi`.

Line ?? is also well-formed because implicit conversion from `simd<T, simd_abi::fixed_size<N>>` to `simd<U, Abi>` is allowed whenever `N == simd<U, Abi>::size()`.

7.8

NATIVE HANDLE

The presence of a `native_handle` function for accessing an internal data member such as e.g. a vector builtin or SIMD intrinsic type is seen as an important feature for adoption in the target communities. Without such a handle the user is constrained to work within the (limited) API defined by the standard. Many SIMD instruction sets have domain-specific instructions that will not easily be usable (if at all) via the standardized interface. A user considering whether to use `simd` or a SIMD extension such as vector builtins or SIMD intrinsics might decide against `simd` just for fear of not being able to access all functionality.¹

I would be happy to settle on an alternative to exposing an lvalue reference to a data member. Consider implementation-defined support casting (`static_cast?`) between `simd` and non-standard SIMD extension types. My understanding is that there could not be any normative wording about such a feature. However, I think it could be useful to add a non-normative note about making `static_cast(?)` able to convert between such non-standard extensions and `simd`.

Guidance from SG1 at Oulu 2016:

Poll: Keep `native_handle` in the wording?

SF	F	N	A	SA
0	6	3	3	0

7.9

LOAD & STORE FLAGS

SIMD loads and stores require at least an alignment option. This is in contrast to implicit loads and stores present in C++, where alignment is always assumed. Many SIMD instruction sets allow more options, though:

- Streaming, non-temporal loads and stores
- Software prefetching

¹ Whether that's a reasonable fear is a different discussion.

In the Vc library I have added these as options in the load store flag parameter of the `load` and `store` functions. However, non-temporal loads & stores and prefetching are also useful for the existing builtin types. I would like guidance on this question: should the general direction be to stick to *only* alignment options for `simd` loads and stores?

The other question is on the default of the load and store flags. Some argue for setting the default to `aligned`, as that's what the user should always aim for and is most efficient. Others argue for `unaligned` since this is safe per default. The Vc library before version 1.0 used aligned loads and stores per default. After the guidance from SG1 I changed the default to unaligned loads and stores with the Vc 1.0 release. Changing the default is probably the worst that could be done, though.² For Vc 2.0 I will drop the default.

For `simd` I prefer no default:

- This makes it obvious that the API has the alignment option. Users should not just take the default and think no more of it.
- If we decide to keep the load constructor, the alignment parameter (without default) nicely disambiguates the load from the broadcast.
- The right default would be application/domain/experience specific.
- Users can write their own load/store wrapper functions that implement their chosen default.

Guidance from SG1 at Oulu 2016:

Poll: Should the interface provide a way to specify a number for over-alignment?

SF	F	N	A	SA
2	6	5	0	0

Poll: Should loads and stores have a default load/store flag?

SF	F	N	A	SA
0	0	7	4	1

The discussion made it clear that we only want to support alignment flags in the load and store operations. The other functionality is orthogonal.

7.10

UNARY MINUS RETURN TYPE

The return type of `simd<T, Abi>::operator-()` is `simd<T, Abi>`. This is slightly different to the behavior of the underlying element type `T`, if `T` is an integral type of

² As I realized too late.

lower integer conversion rank than `int`. In this case integral promotion promotes the type to `int` before applying unary minus. Thus, the expression `-T()` is of type `int` for all `T` with lower integer conversion rank than `int`. This is widening of the element size is likely unintended for SIMD vector types.

Fundamental types with integer conversion rank greater than `int` are not promoted and thus a unary minus expression has unchanged type. This behavior is copied to element types of lower integer conversion rank for `simd`.

There may be one interesting alternative to pursue here: We can make it ill-formed to apply unary minus to unsigned integral types. Anyone who wants to have the modulo behavior of a unary minus could still write `0u - x`.

7.11

MAX_FIXED_SIZE

In Kona, LEWG asked why `max_fixed_size` is not dependent on `T`. After some consideration I am convinced that the correct solution is to make `max_fixed_size` a variable template, dependent on `T`.

The reason to restrict the number of elements N in a fixed-size `simd` type at all, is to inhibit misuse of the feature. The intended use of the fixed-size ABI, is to work with a number of elements that is somewhere in the region of the number of elements that can be processed efficiently concurrently in hardware. Implementations may want to use recursion to implement the fixed-size ABI. While such an implementation can, in theory, scale to any N , experience shows that compiler memory usage and compile times grow significantly for “too large” N . The optimizer also has a hard time to optimize register / stack allocation optimally if N becomes “too large”. Unsuspecting users might not think of such issues and try to map their complete problem to a single `simd` object. Allowing implementations to restrict N to a value that they can and want to support thus is useful for users and implementations. The value itself should not be prescribed by the standard as it is really just a QoI issue.

However, why should the user be able to query the maximum N supported by the implementation?

- In principle, a user can always determine the number using SFINAE to find the maximum N that he can still instantiate without substitution failure. Not providing the number thus provides no “safety” against “bad usage”.
- A developer may want to use the value to document assumptions / requirements about the implementation, e.g. with a static assertion.
- A developer may want to use the value to make code portable between implementations that use a different `max_fixed_size`.

Making the `max_fixed_size` dependent on `T` makes sense because most hardware can process a different number of elements in parallel depending on `T`. Thus, if an implementation wants to restrict N to some sensible multiple of the hardware capabilities, the number must be dependent on `T`.

In Kona, LEWG also asked whether there should be a provision in the standard to ensure that a native `simd` of 8-bit element type is convertible to a fixed-size `simd` of 64-bit element type. It was already there ([`simd.abi`]: “for every supported `simd<T, A>` (see [`simd.overview`]), where `A` is an implementation-defined ABI tag, $N = \text{simd}<T, A>::\text{size}()$ must be supported”). Note that this does not place a lower bound on `max_fixed_size`. The wording allows implementations to support values for fixed-size `simd` that are larger than `max_fixed_size`. I.e. $N \leq \text{max_fixed_size}$ works; whether $N > \text{max_fixed_size}$ works is unspecified.

A

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B

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